

Suv bilan ishlaydigan sholi keli (Water-powered rice-pounding mortar)

A. Kirish (Introduction)

Guruch (rice) Vetnamda ko'pchilik aholining asosiy oziq-ovqat mahsulotidir. Palla guruchdan (paddy rice) oq guruch olish uchun qobiqni (husk) ajratish ("hulling" – qobiqni ajratish) va kepak qatlamini (bran layer) ajratish ("milling" – maydalash) kerak. Shimoliy Vetnamning tog'li hududlari suv oqimlariga boy va u yerda yashovchi odamlar kepak qatlamini ajratish uchun suv bilan ishlaydigan sholi keli (water-powered rice-pounding mortar) dan foydalanadilar. 1-rasmda (Rasm (1).png) bunday kellardan biri, 2-rasmda (Rasm (2).png) uning ishlash prinsipi ko'rsatilgan.

B. Konstruksiya va ishlash (Design and operation)

1. Konstruksiya (Design)

1-rasmda ko'rsatilgan sholi kelinig quyidagi qismlari bor:

- Keli (mortar) – asosan guruch uchun yog'och idish.
- Richag (lever) – bir uchi kattaroq, bir uchi kichikroq bo'lgan daraxt tanasi. U gorizontal o'q atrofida aylana oladi. Kichik uchiga (pestle – keli to'qmog'i) richagga perpendikulyar ravishda biriktirilgan. To'qmoqning uzunligi shundayki, richag gorizontal yotganda u keli ichidagi guruchga tegadi. Richagning katta uchi chelak (bucket) hosil qilish uchun o'yilgan. Chelakning shakli kelinig ishlashi uchun juda muhim.

2. Ishlash rejimlari (Modes of operation)

Keli ikki rejimga ega.

Ishchi rejim (Working mode) Bu rejimda keli 2-rasmda (Rasm (2).png) tasvirlangan operatsion sikl (operation cycle) dan o'tadi.

Sholi urish funksiyasi (rice-pounding function) 2-rasmning f) bosqichida to'qmoqdan guruchga o'tkazilgan ish natijasida keladi. Agar biron sababga ko'ra to'qmoq hech qachon guruchga tegmasa, biz keli ishlamayapti (not working) deymiz.

Richag ko'tarilgan holdagi tinch rejim (Rest mode with the lever lifted up)

Operatsion siklining c) bosqichida (2-rasm), qiyalik burchagi α ortgan sari chelakdagi suv miqdori kamayadi. Vaqtning ma'lum bir momentida suv miqdori richagning og'irligini muvozanatlash uchun yetarli bo'ladi. Bu paytdagi qiyalik burchagini β deb belgilaymiz. Agar richag β burchakda ushlab turilsa va boshlang'ich burchak tezligi nol bo'lsa, u holda richag abadiy shu holatda qoladi. Bu richag ko'tarilgan holdagi tinch rejimdir. Bu pozitsiyaning barqarorligi (stability) chelakka suv kelish tezligiga (flow rate) Φ bog'liq. Agar Φ ba'zi Φ_2 qiymatidan oshsa, bu tinch rejim barqaror bo'ladi va keli ishchi rejimda bo'la olmaydi.

Boshqacha aytganda, Φ_2 – keli ishlamasligi uchun minimal oqim tezligi (minimal flow rate for the mortar not to work).



1-rasm: Suv bilan ishlaydigan sholi keli (Water-powered rice-pounding mortar).

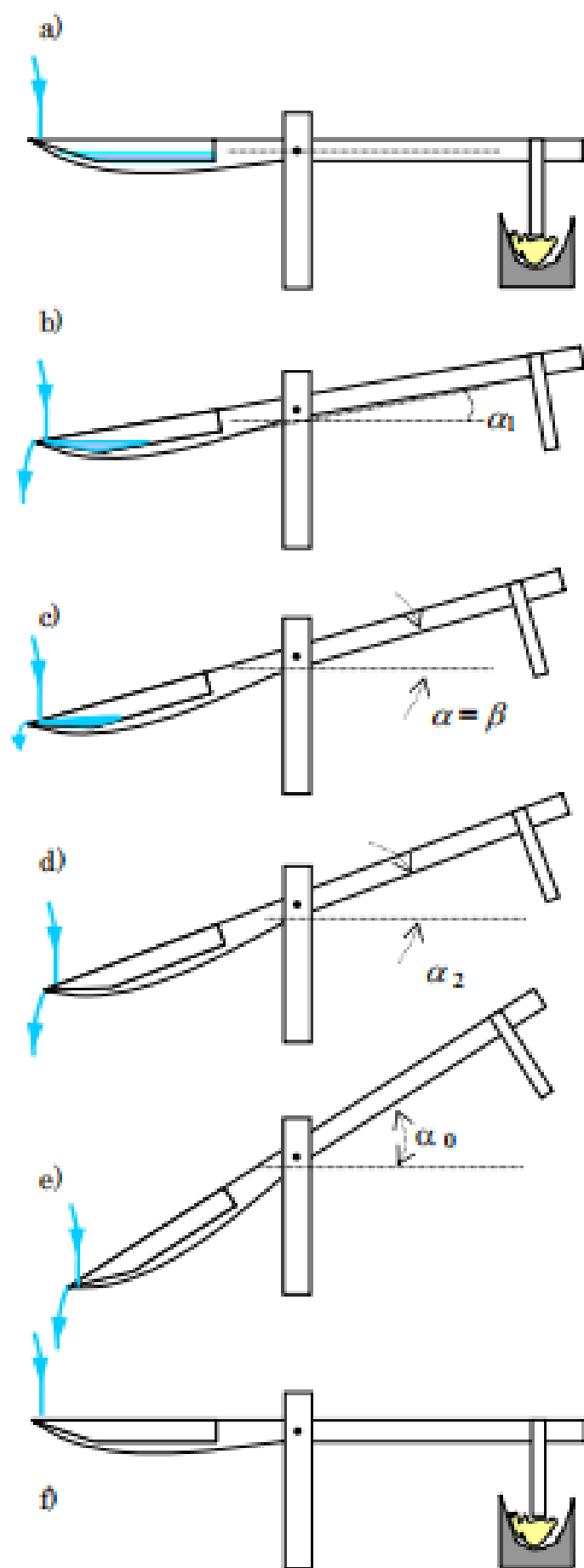


Figure 2

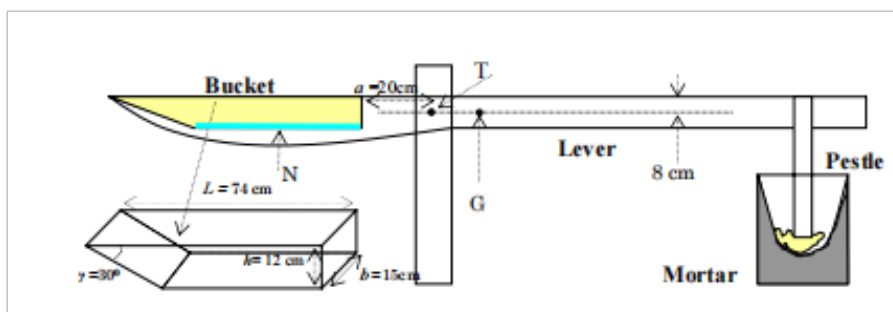
2-rasm: Suv bilan ishlaydigan sholi keling ishlash sikli (Operation cycle).

C. Masala (The problem)

Quyidagi parametrlarga ega suv bilan ishlaydigan sholi kelini ko'rib chiqaylik (3-rasm – Rasm (3).png):

- Richagning massasi (suvsiz, ammo to'qmoq bilan birga): $M = 30 \text{ kg}$.
- Richagning massa markazi (center of mass) G . Richag T o'qi atrofida aylanadi (rasmda T nuqtasiga proyeksiyalangan).
- Richagning T o'qiga nisbatan inersiya momenti (moment of inertia): $I = 12 \text{ kg} \cdot \text{m}^2$.
- Chelakda suv bo'lganda, suv massasi m , suvning massa markazi N bilan belgilanadi.
- Richagning gorizontaal o'qqa nisbatan qiyalik burchagi (tilt angle): α .
- Asosiy uzunlik o'lchovlari 3-rasmda keltirilgan.

Aylanma o'qidagi ishqalanishni (friction) va chelakka tushayotgan suvning kuchini e'tiborsiz qoldiring. Ushbu masalada biz suv sathi doimo gorizontaal deb yaqinlashish (approximation) qilamiz.



3-rasm: Sholi keling konstruksiyasi va o'lchamlari.

1. Kelining tuzilishi (The structure of the mortar)

Boshlang'ichda chelak bo'sh va richag gorizontaal yotadi. So'ngra chelakka suv to'la boshlaydi, to richag aylana boshlaguncha. Bu paytda chelakdagi suv miqdori $m = 1.0 \text{ kg}$.

- 1.1 Richagning massa markazi G dan aylanish o'qi T gacha bo'lgan masofani (distance) aniqlang. Ma'lumki, chelak bo'sh bo'lganda GT gorizontaal (horizontal) bo'ladi.
- 1.2 Richag va gorizontaal o'q orasidagi burchak α_1 ga yetganda chelakdan suv chiqa boshlaydi (starts flowing out). Bu burchak α_2 bo'lganda chelak butunlay bo'shaydi. α_1 va α_2 ni aniqlang.
- 1.3 $\mu(\alpha)$ – richag va chelakdagi suvning og'irligidan kelib chiqadigan umumiy moment (total torque) (T o'qiga nisbatan) bo'lsin. $\mu(\alpha)$ $\alpha = \beta$ da nolga teng. β ni va shu paytdagi chelakdagi suv massasi m_1 ni aniqlang.

2. Ishchi rejim parametrlari (Parameters of the working mode)

Suv chelakka o'zgaras Φ oqim tezligi (flow rate) bilan kelsin. Richag harakatlantayotganda chelakka kirayotgan suv miqdori ahamiyatsiz. Ushbu qismda ishchi sikl davomida inersiya momentining o'zgarishini e'tiborsiz qoldiring.

- 2.1 Bir ishchi sikl davomida μ momentning α burchakka bog'liqlik grafigi $\mu(\alpha)$ ni chizing. α_1 , α_2 va $\alpha = 0$ burchaklaridagi $\mu(\alpha)$ qiymatlarini aniq yozing.
- 2.2 2.1 bo'limda topilgan grafikdan foydalanib, $\mu(\alpha)$ hosil qilgan umumiy energiya W_{total} va to'qmoqdan guruchga o'tkazilgan ish W_{pounding} ning geometrik talqinini (geometric interpretation) muhokama qiling.
- 2.3 μ ning α ga nisbatan grafigidan foydalanib, α_0 va W_{pounding} ni baholang (chelakka kirayotgan va chiqayotgan suvning kinetik energiyasini ahamiyatsiz deb hisoblang). Agar hisoblashni soddalashtirsa, egri chiziqlarni zigzag chiziqlar bilan almashtirishingiz mumkin.

3. Tinch rejim (The rest mode)

Suv chelakka o'zgaras Φ tezlik bilan kelsin, ammo richag harakati davomida chelakka kirayotgan suv miqdorini e'tiborsiz qoldirib bo'lmaydi.

- 3.1 Chelak doim to'lib toshib turishi (always overflowing) faraz qiling.
 - 3.1.1 $\alpha = \beta$ atrofida μ momentning α ga bog'liqlik grafigini chizing. Richagning $\alpha = \beta$ pozitsiyasi qanday muvozanat turiga (kind of equilibrium) kiradi?
 - 3.1.2 $\alpha = \beta + \Delta\alpha$ bo'lganda, $\Delta\alpha$ kichik deb, $\mu(\alpha)$ momentning analitik ko'rinishini $\Delta\alpha$ orqali toping.
 - 3.1.3 Richagning harakat tenglamasini (equation of motion) yozing, u $\alpha = \beta + \Delta\alpha$ holatidan ($\Delta\alpha$ kichik) boshlang'ich tezligi nol bilan harakatlanadi. Harakat yaxshi aniqlikda garmonik tebranish (harmonic oscillation) ekanligini ko'rsating. τ davrini (period) hisoblang.
- 3.2 Berilgan Φ da, chelak faqat richag yetarlicha sekin harakat qilganda har doim to'lib toshadi. Garmonik tebranish amplitudasining (amplitude) yuqori chegarasi mavjud, u Φ ga bog'liq. Richag amplitudasi 1° bo'lgan garmonik tebranma harakat qila olishi uchun Φ ning minimal qiymati Φ_1 ni (kg/s da) aniqlang.
- 3.3 Faraz qilaylik, Φ yetarlicha katta, shuning uchun richag qiyalik burchagi α_2 dan α_1 gacha kamayadigan erkin harakat davrida chelak doimo to'lib toshgan bo'ladi. Biroq, agar Φ juda katta bo'lsa, keli ishlay olmaydi. Richag harakati garmonik tebranish deb faraz qilib, sholi keling ishlamasligi uchun minimal oqim tezligi Φ_2 ni baholang.

Cherenkov nuri va halqaviy tasvir hisoblagichi

Yorug'lik vakuumda c tezlik bilan tarqaladi. Hech qanday zarra c dan katta tezlik bilan harakatlana olmaydi. Biroq, shaffof muhitda (transparent medium) zarra shu muhitdagi yorug'lik tezligi $\frac{c}{n}$ dan katta v tezlik bilan harakatlanishi mumkin, bu yerda n – muhitning

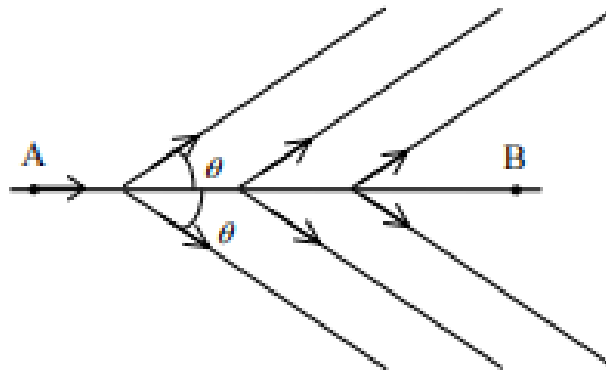
sindirish korsatkichi (refraction index). Tajriba (Cherenkov, 1934) va nazariya (Tamm va Frank, 1937) shuni korsatdiki, zaryadlangan zarra (charged particle) sindirish korsatkichi n bolgan shaffof muhitda $v > \frac{c}{n}$ tezlik bilan harakatlanganda, Cherenkov nuri (Cherenkov light) deb ataladigan nurlanish chiqaradi. Bu nurlanish traektoriya (trajectory) bilan $\theta = \arccos \frac{1}{\beta n}$ (1) burchak hosil qilgan yonalishlarda tarqaladi, bu yerda $\beta = \frac{v}{c}$.

1. Ushbu faktni aniqlash (To establish this fact)

Zarra togri chiziq boylab ozgarmas $v > \frac{c}{n}$ tezlik bilan harakatlanayotganini korib chiqaylik. U A nuqtadan 0 vaqtda, B nuqtadan t_1 vaqtda otadi. Masala AB atrofida aylanishga nisbatan simmetrik bolgani uchun AB ni oz ichiga olgan tekislikdagi yoruglik nurlarini korib chiqish yetarli.

A va B orasidagi istalgan C nuqtada zarra sferik yoruglik tolqini (spherical light wave) chiqaradi, u $\frac{c}{n}$ tezlik bilan tarqaladi. Berilgan t vaqtdagi tolqin fronti (wave front) deb shu vaqtda barcha shu sferalarning konverti (envelope) ga aytiladi.

- 1.1 t_1 vaqtidagi tolqin frontini aniqlang va uning zarra traektoriyasini oz ichiga olgan tekislik bilan kesishmasini (intersection) chizing.



Tolqin fronti va zarra traektoriyasi kesishmasi.

- 1.2 Ushbu kesishma va zarra traektoriyasi orasidagi φ burchakni n va β orqali ifodalang.

2.

$v > \frac{c}{n}$ tezlik bilan harakatlanuvchi zarralar dastasi (beam) togri IS chiziq boylab, θ burchak kichik bolgan holda harakatlanadi. Dasta fokus masofasi (focal length) f va markazi C bolgan botiq sferik kozguga (concave spherical mirror) S nuqtada keladi. SC SI bilan kichik α burchak hosil qiladi (javoblar varagidagi rasmga qarang). Zarra dastasi kozguning fokal tekisligida (focal plane) halqaviy tasvir (ring image) hosil qiladi.

Ushbu faktni tushuntirib, chizma yordamida korsating. Halqaning markazi O va radiusi r ning joylashuvini bering.

Ushbu qurilma *halqaviy Cherenkov hisoblagichlarida* (ring imaging Cherenkov counters – RICH) ishlatiladi va zarra otadigan muhit *radiator* (radiator) deb ataladi.

Eslatma: masalaning barcha savollarida α va θ dagi ikkinchi va undan yuqori tartibli hadlar etiborsiz qoldiriladi.

3.

Malum impulsi (momentum) $p = 10.0 \text{ GeV}/c$ bolgan zarra dastasi uch turdagi zarralardan iborat: protonlar (protons), kaonlar (kaons) va pionlar (pions), ularning tinch massasi (rest mass) mos ravishda $M_p = 0.94 \text{ GeV}/c^2$, $M_\kappa = 0.50 \text{ GeV}/c^2$ va $M_\pi = 0.14 \text{ GeV}/c^2$. Eslatma: pc va Mc^2 energiya olchovligiga ega, 1 eV – elektronni 1 V kuchlanish bilan tezlatganda oladigan energiya, $1 \text{ GeV} = 10^9 \text{ eV}$, $1 \text{ MeV} = 10^6 \text{ eV}$.

Zarralar dastasi havo muhitidan (radiator) P bosim (pressure) ostida otadi. Havoning sindirish korsatkichi havo bosimiga $n = 1 + aP$ munosabat bilan bogliq, bu yerda $a = 2.7 \times 10^{-4} \text{ atm}^{-1}$.

- 3.1 Har uch zarra turi uchun ular Cherenkov nuri chiqarishi uchun havo bosimining minimal qiymati $P_{\min}$ ni hisoblang.
- 3.2 Kaonlarning halqa tasviri radiusi pionlarnikining yarmiga teng boladigan bosim P_1 ni hisoblang. Bu holda θ_κ va θ_π qiymatlarini hisoblang. Ushbu bosimda protonlarning halqa tasvirini kuzatish mumkinmi?

4.

Endi dasta mukammal monoxromatik (perfectly monochromatic) emas deb faraz qilaylik: zarralar momenti $10 \text{ GeV}/c$ markazida joylashgan, yarim balandlikdagi yarim kengligi (half width at half height) Δp ga teng oraliqda taqsimlangan. Bu halqa tasvirining kengayishiga olib keladi, mos ravishda θ taqsimoti yarim balandlikdagi yarim kengligi $\Delta\theta$ ga ega. Radiatorning bosimi 3.2 da aniqlangan P_1 ga teng.

- 4.1 Pionlar va kaonlar holatida $\frac{\Delta\theta}{\Delta p}$ ning qiymatlari $\frac{\Delta\theta_\kappa}{\Delta p}$ va $\frac{\Delta\theta_\pi}{\Delta p}$ ni hisoblang.
- 4.2 Ikki halqa tasviri orasidagi farq $\theta_\pi - \theta_\kappa$ yarim kengliklar yigindisidan 10 marta katta bolganda, yani $\theta_\pi - \theta_\kappa > 10\Delta\theta$ (bu yerda $\Delta\theta = \Delta\theta_\kappa + \Delta\theta_\pi$), ikki halqa tasvirini yaxshi ajratish mumkin. Ikki halqa tasvirini hali yaxshi ajratish mumkin bolgan Δp ning maksimal qiymatini hisoblang.

5.

Cherenkov bu effektini radioaktiv manba yonida joylashgan bir shisha suvni kuzatayotganda kashf etdi. U suv ichidagi suvning nur chiqarayotganini kordi.

- 5.1 Tinch massasi M bolgan zarra suvda harakatlanib, Cherenkov nuri chiqarishi uchun uning minimal kinetik energiyasi (minimal kinetic energy) $T_{\min}$ ni toping. Suvning sindirish korsatkichi $n = 1.33$.
- 5.2 Cherenkov ishlatgan radioaktiv manba tinch massasi $M_\alpha = 3.8 \text{ GeV}/c^2$ bolgan α zarralari (geliy yadrolari – helium nuclei) yoki tinch massasi $M_e = 0.51 \text{ MeV}/c^2$ bolgan β zarralari (elektronlar – electrons) chiqaradi. α va β zarralari uchun $T_{\min}$ ning son qiymatlarini hisoblang. Radioaktiv manbalar chiqaradigan zarralarning kinetik energiyasi hech qachon bir necha MeV dan oshmasligini bilgan holda, Cherenkov kuzatgan nurlanishga qaysi zarralar sabab bolganini aniqlang.

6.

Masalaning oldingi qismlarida Cherenkov effektining λ tolqin uzunligiga (wavelength) bogliqligi etiborsiz qoldirilgan. Endi zarra Cherenkov nurlanishining korinadigan diapazonni (visible range) (tolqin uzunliklari $0.4 \mu\text{m}$ dan $0.8 \mu\text{m}$ gacha) oz ichiga olgan keng uzluksiz spektrga (broad continuous spectrum) ega ekanligini hisobga olamiz. Shuningdek, radiatorning sindirish korsatkichi n bu diapazonda λ ortishi bilan $n - 1$ ning 2% ga chiziqli ravishda kamayishini bilamiz.

- 6.1 6 atm bosimdagi havoda $10.0 \text{ GeV}/c$ aniq momentga ega bolgan pionlar dastasini korib chiqaylik. Korinadigan diapazonning ikki chegarasi bilan bogliq burchak farqi $\delta\theta$ ni toping.
- 6.2 Shu asosda, momentumi $p = 10 \text{ GeV}/c$ markazida joylashgan va yarim balandlikdagi yarim kengligi $\Delta p = 0.3 \text{ GeV}/c$ ga teng oraliqda taqsimlangan pionlar dastasining halqa tasviriga dispersiyaning (dispersion) tasirini sifat jihatidan organing.
 - 6.2.1 Dispersiya (sindirish korsatkichining ozgarishi) tufayli kengayish va dastaning axromatikligi (momentumning ozgarishi) tufayli kengayishni hisoblang.
 - 6.2.2 Javoblar varagida tegishli katakchalarni belgilab, halqa rangining (color) uning ichki chetidan tashqi chetiga otishda qanday ozgarishini tasvirlang.

Havo haroratining balandlik bilan ozgarishi, atmosfera barqarorligi va havo ifloslanishi (CHANGE OF AIR TEMPERATURE WITH ALTITUDE, ATMOSPHERIC STABILITY AND AIR POLLUTION)

Vertikal havo harakati (vertical motion of air) bulutlar va yogingarchiliklarning shakllanishi hamda havo ifloslantiruvchi moddalarning tarqalishi (dispersal of air pollutants) kabi koplal atmosfera jarayonlarini boshqaradi. Agar atmosfera barqaror (stable) bolsa, vertikal harakat cheklangan boladi va ifloslantiruvchi moddalar tarqalish va suyulish (dispersed and diluted) orniga emissiya manbai atrofida toplanadi. Shu bilan birga, beqaror (unstable) atmosferada havoning vertikal harakati ifloslantiruvchi moddalarning vertikal tarqalishini ragbatlantiradi. Shuning uchun ifloslantiruvchi moddalarning kontsentratsiyasi (concentration) nafaqat emissiya manbalarining kuchiga, balki atmosferaning barqarorligiga ham bogliq.

Atmosfera barqarorligini meteorologiyadagi havo zarrasi (air parcel) tushunchasidan foydalanib aniqlaymiz va atmosferada adiabatik ravishda kotarilayotgan yoki tushayotgan havo zarrasining haroratini atrofda havo harorati bilan solishtiramiz. Kop hollarda ifloslantiruvchi moddalarni oz ichiga olgan va yerdan kotarilayotgan havo zarrasi malum bir balandlikda, aralashma balandligi (mixing height) deb ataladigan balandlikda toxtab qolishini koramiz. Aralashma balandligi qancha katta bolsa, havo ifloslantiruvchi moddalar kontsentratsiyasi shuncha past boladi. Biz Hanoi metropoliten hududida ertalabki soatlarda mototsikllar chiqaradigan uglerod oksidi (CO) kontsentratsiyasini va aralashma balandligini baholaymiz, bunda 119 m dan yuqori balandliklarda harorat inversiyasi (temperature inversion – harorat balandlik bilan ortadi) tufayli vertikal aralashish cheklangan.

Havoni ideal ikki atomli gaz (ideal diatomic gas) deb hisoblaymiz, molyar massasi (molar mass) $\mu = 29 \text{ g/mol}$.

Kvazi muvozanatli (quasi equilibrium) adiabatik ozgarish $pV^\gamma = \text{const}$ tenglamasiga boysunadi, bu yerda

$$\gamma = \frac{c_p}{c_v}$$

gazning izobarik va izoxorik issiqlik sigimlari (isobaric and isochoric heat capacities) nisbati.

Talaba kerak bolganda quyidagi malumotlardan foydalanishi mumkin:

- Universal gaz doimiysi (universal gas constant): $R = 8.31 \text{ J}/(\text{mol} \cdot \text{K})$.
- Yerdagi atmosfera bosimi (atmospheric pressure on ground): $p_0 = 101.3 \text{ kPa}$.
- Erkin tushish tezlanishi (acceleration due to gravity) o'zgaras: $g = 9.81 \text{ m/s}^2$.
- Havo uchun molyar izobarik issiqlik sig'imi (molar isobaric heat capacity): $c_p = \frac{7}{2}R$.
- Havo uchun molyar izoxorik issiqlik sig'imi (molar isochoric heat capacity): $c_v = \frac{5}{2}R$.

Matematik ko'rsatmalar (Mathematical hints)

1. $\int \frac{dx}{A+Bx} = \frac{1}{B} \int \frac{d(A+Bx)}{A+Bx} = \frac{1}{B} \ln(A+Bx)$
2. Differensial tenglamaning yechimi $\frac{dx}{dt} + Ax = B$ (bu yerda A va B o'zgaras) quyidagicha: $x(t) = x_1(t) + \frac{B}{A}$, bu yerda $x_1(t) - \frac{dx}{dt} + Ax = 0$ tenglamasining yechimi.
3. $\lim_{x \rightarrow \infty} \left(1 + \frac{1}{x}\right)^x = e$

1. Bosimning balandlik bilan o'zgarishi (Change of pressure with altitude)

- 1.1. Atmosfera harorati bir xil (uniform) va T_0 ga teng deb faraz qiling. Atmosfera bosimi p ni balandlik z funksiyasi sifatida ifodalovchi ifodani yozing.
- 1.2. Atmosfera harorati balandlik bilan $T(z) = T(0) - \Lambda z$ munosabatga ko'ra o'zgaradi deb faraz qiling, bu yerda Λ – o'zgaras, atmosferaning harorat pasayish tezligi (temperature lapse rate) deyiladi (haroratning vertikal gradienti $-\Lambda$).
 - 1.2.1. Atmosfera bosimi p ni balandlik z funksiyasi sifatida ifodalovchi ifodani yozing.
 - 1.2.2. Erkin konvektsiya (free convection) deb ataladigan jarayon havo zichligi (air density) balandlik bilan ortganida sodir bo'ladi. Qaysi Λ qiymatlarida erkin konvektsiya sodir bo'ladi?

2. Vertikal harakatdagi havo zarrasi haroratining o'zgarishi (Change of the temperature of an air parcel in vertical motion)

Atmosferada yuqoriga va pastga harakatlanuvchi havo zarrasini (air parcel) ko'rib chiqaylik. Havo zarrasi – bir necha metr kenglikdagi etarlicha katta, mustaqil termodinamik ob'ekt sifatida qaraladigan, lekin harorati bir xil deb hisoblash uchun yetarlicha kichik bo'lgan havo tanasidir. Havo zarrasining vertikal harakati kvazi-adiabatik jarayon (quasi adiabatic process) sifatida qaralishi mumkin, ya'ni atrofdagi havo bilan issiqlik almashinuvi ahamiyatsiz. Agar havo zarrasi atmosferada ko'tarilsa, u kengayadi (expands) va soviydi (cools). Aksincha, agar u pastga harakatlansa, tashqi bosim ortishi zarra ichidagi havoni siqadi (compresses) va uning harorati ortadi.

Zarra o'lchami katta bo'lmagani uchun, zarra chegarasidagi turli nuqtalarda atmosfera bosimi bir xil $p(z)$ qiymatga ega deb hisoblash mumkin, bu yerda z – zarra markazining balandligi. Zarradagi harorat bir xil va $T_{\text{parcel}}(z)$ ga teng, bu odatda atrofdagi havoning harorati $T(z)$ dan farq qiladi. 2.1 va 2.2 qismlarida $T(z)$ ning shakli haqida hech qanday faraz qilmaymiz.

- 2.1. Zarra haroratining T_{parcel} balandlik bilan o'zgarishi $\frac{dT_{\text{parcel}}}{dz} = -G$ bilan aniqlanadi. $G(T, T_{\text{parcel}})$ ning ifodasini keltirib chiqaring.
- 2.2. Atmosferaning ixtisoslashgan shartini ko'rib chiqaylik, bunda har qanday z balandlikda atmosfera harorati T zarra haroratiga teng: $T(z) = T_{\text{parcel}}(z)$. G ning $T = T_{\text{parcel}}$ bo'lgan paytdagi qiymatini Γ bilan belgilaymiz, ya'ni $\Gamma = -\frac{dT_{\text{parcel}}}{dz}$ (bunda $T = T_{\text{parcel}}$). Γ quruq adiabatik pasayish tezligi (dry adiabatic lapse rate) deb ataladi.
 - 2.2.1. Γ ifodasini keltirib chiqaring.
 - 2.2.2. Γ ning son qiymatini hisoblang.
 - 2.2.3. Atmosfera haroratining $T(z)$ balandlik funksiyasi sifatidagi ifodasini keltirib chiqaring.
- 2.3. Atmosfera harorati $T(z) = T(0) - \Lambda z$ (bu yerda Λ – o'zgarmas) munosabat bilan berilgan deb faraz qiling. Zarra haroratining $T_{\text{parcel}}(z)$ balandlikka bog'liqligini toping.
- 2.4. $|\Lambda z| \ll T(0)$ va $T(0) \approx T_{\text{parcel}}(0)$ bo'lgan holda $T_{\text{parcel}}(z)$ ning taqribiy ifodasini yozing.

3. Atmosfera barqarorligi (The atmospheric stability)

Ushbu qismda T balandlik bilan chiziqli o'zgaradi deb faraz qilamiz.

- 3.1. Dastlab z_0 balandlikda atrofdagi havo bilan muvozanatda bo'lgan havo zarrasini ko'rib chiqaylik, ya'ni uning harorati atrofdagi havo harorati $T(z_0)$ ga teng. Agar zarra biroz yuqoriga va pastga (masalan, atmosfera turbulentsiyasi tufayli) siljigan bo'lsa, quyidagi uch holatdan biri sodir bo'lishi mumkin:
 - Havo zarrasi dastlabki z_0 balandlikka qaytadi, zarra muvozanati barqaror (stable). Atmosfera barqaror deyiladi.

- Zarra dastlabki yo'nalishida harakatni davom ettiradi, zarra muvozanati beqaror (unstable). Atmosfera beqaror.
- Havo zarrasi yangi holatida qoladi, zarra muvozanati indiferent (neutral). Atmosfera neytral deyiladi.

Atmosferaning barqaror, beqaror yoki neytral bo'lishi uchun Λ ga qanday shart qo'yiladi?

- 3.2. Bir zarra erdagi harorati $T_{\text{parcel}}(0)$ atrofdagi havo harorati $T(0)$ dan yuqori. Arximed kuchi (buoyancy force) zarra ko'tarilishiga sabab bo'ladi. Barqaror atmosfera holatida zarra erisha oladigan maksimal balandlikning ifodasini Λ va Γ orqali keltirib chiqaring.

4. Aralashma balandligi (The mixing height)

- 4.1. 1-jadval (Table 1) da Hanoi shahrida noyabr oyining bir kuni ertalab soat 7:00 da radiozond (radio sounding balloon) tomonidan qayd etilgan havo haroratlari ko'rsatilgan. Haroratning balandlik bilan o'zgarishini taxminan $T(z) = T(0) - \Lambda z$ formulasi bilan uch qatlamda ($0 < z < 96 \text{ m}$, $96 \text{ m} < z < 119 \text{ m}$ va $119 \text{ m} < z < 215 \text{ m}$) turlicha Λ pasayish tezliklari bilan tasvirlash mumkin.

Erdagi harorati $T_{\text{parcel}}(0) = 22^\circ\text{C}$ bo'lgan havo zarrasini ko'tarilayotgan deb hisoblang. 1-jadvaldagi ma'lumotlar va yuqoridagi chiziqli yaqinlashishga asoslanib, zarra haroratini 96 m va 119 m balandliklarda hisoblang.

- 4.2. Zarra erisha oladigan maksimal balandlik H va zarra harorati $T_{\text{parcel}}(H)$ ni aniqlang. H aralashma balandligi (mixing height) deb ataladi. Erdagi ifloslantiruvchi moddalar (air pollutants) atmosferadagi havo bilan aralashishi (masalan, shamol, turbulentsiya va dispersiya) va shu qatlam ichida suyultirilishi mumkin.

Table 1: 1-jadval (Table 1) – Radiozond tomonidan Hanoi shahrida noyabr oyining bir kuni ertalab soat 7:00 da qayd etilgan ma'lumotlar.

Balandlik (Altitude), m	Harorat (Temperature), °C
5	21.5
60	20.6
64	20.5
69	20.5
75	20.4
81	20.3
90	20.2
96	20.1
102	20.1
109	20.1
113	20.1
119	20.1
128	20.2
136	20.3
145	20.4
153	20.5
159	20.6
168	20.8
178	21.0
189	21.5
202	21.8
215	22.0
225	22.1
234	22.2
246	22.3
257	22.3

5. Hanoi shahrida ertalabki mototsikl soatlarida uglerod oksidi (CO) ifloslanishini baholash (Estimation of carbon monoxide (CO) pollution during a morning motorbike rush hour in Hanoi)



Hanoi metropoliten hududini rasmda ko'rsatilganidek, asosi L va W bo'lgan to'rtburchak (rectangle) bilan yaqinlashtirish mumkin, bir tomoni Qizil daryoning (Red River) janubig'arbiy qirg'og'i bo'ylab joylashgan.

Ertalabki soatlarda (soat 7:00 dan 8:00 gacha) yo'lda 8×10^5 mototsikl borligi, har biri o'rtacha 5 km masofa bosib o'tishi va har bir kilometrda 12 g CO chiqarishi taxmin qilinadi. CO ifloslantiruvchi moddaning miqdori taxminan rush hour davomida vaqt bo'yicha bir tekis, o'zgarmas tezlik M da chiqariladi deb hisoblanadi. Shu bilan birga, toza shimoli-sharq shamoli (clean north-east wind) Qizil daryoga perpendikulyar (ya'ni to'rtburchakning L tomonlariga perpendikulyar) ravishda u tezlik bilan esadi, shahar ustidan xuddi shu tezlikda o'tadi va CO ifloslangan havoning bir qismini shahar atmosferasidan tashqariga olib chiqadi.

Shuningdek, quyidagi taxminiy modeldan foydalanamiz:

- CO aralashma qatlami (mixing layer) hajmi (volume) bo'ylab tez tarqaladi, shuning uchun t vaqtidagi CO kontsentratsiyasi $C(t)$ o'lchamlari L , W va H bo'lgan to'rtburchak quti (box) ichida o'zgarmas deb hisoblanishi mumkin.
- Qutiga shamol bo'yicha kiruvchi havo (upwind air) toza va shamolga parallel tomonlardan qutidan ifloslanish yo'qotilmaydi deb faraz qilinadi.
- Soat 7:00 dan oldin atmosferadagi CO kontsentratsiyasi ahamiyatsiz.

5.1. CO ifloslantiruvchi kontsentratsiyasi $C(t)$ ning vaqt funksiyasi sifatidagi differensial tenglamasini keltirib chiqaring.

5.2. Ushbu tenglamaning $C(t)$ uchun yechimini yozing.

5.3. Soat 8:00 da $C(t)$ kontsentratsiyasining son qiymatini hisoblang.

Berilgan: $L = 15$ km, $W = 8$ km, $u = 1$ m/s.

DIFFERENTIAL THERMOMETRIC METHOD

Ushbu masalada differensial termometrik usuldan (differential thermometric method) foydalanib, quyidagi ikkita topshiriqni bajariladi:

1. Kristall qattiq moddaning qotish haroratini (temperature of solidification) topish.
2. Quyosh batareyasining (solar cell) samaradorligini (efficiency) aniqlash.

A. Differensial termometrik usul (Differential thermometric method)

Ushbu tajribada haroratni o'lchash uchun oldinga yo'naltirilgan (forward biased) kremniy diodlari (silicon diodes) harorat datchiklari (temperature sensors) sifatida ishlatiladi. Agar diod orqali o'tadigan elektr toki (electric current) o'zgarmas bo'lsa, u holda dioddagi kuchlanish pasayishi (voltage drop) haroratga quyidagi munosabat bilan bog'liq:

$$V(T) = V(T_0) - \alpha(T - T_0) \quad (1)$$

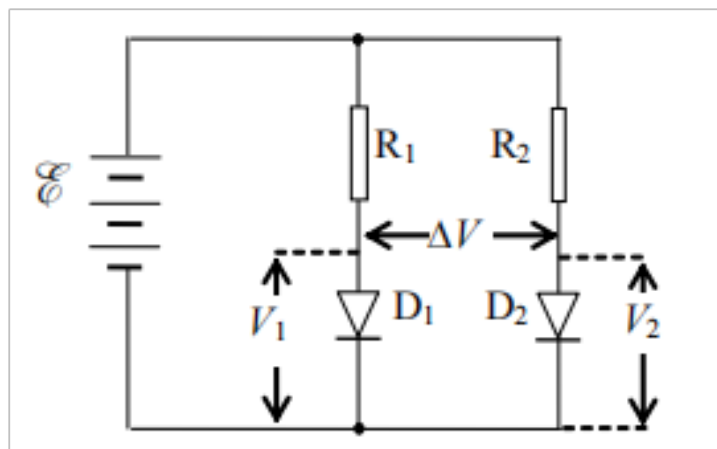
bu yerda $V(T)$ va $V(T_0)$ – mos ravishda dioddagi kuchlanish pasayishi T haroratda va xona harorati T_0 da ($^{\circ}\text{C}$ da o'lchanadi), va koeffitsiyent

$$\alpha = (2.00 \pm 0.03) \text{ mV}/^\circ\text{C} \quad (2)$$

$V(T_0)$ qiymati dioddan diodga biroz farq qilishi mumkin.

Agar ikkita shunday diod turli haroratlarga joylashtirilsa, haroratlar orasidagi farqni ikki dioddagi kuchlanish pasayishlari farqidan o'lchash mumkin. Kuchlanish pasayishlarining farqi, differensial kuchlanish (differential voltage) deb ataladi, yuqori aniqlik bilan o'lchanishi mumkin; shuning uchun harorat farqi ham yuqori aniqlik bilan o'lchanishi mumkin. Bu usul differensial termometrik usul deb ataladi.

Ushbu tajribada diodlar bilan ishlatiladigan elektr zanjiri (electric circuit) 1-rasmda (Rasm (1).png) ko'rsatilgan. D_1 va D_2 diodlari 9V batareya (battery) orqali, 10 k rezistorlar R_1 va R_2 orqali to'g'ri yo'nalishda kuchlanish bilan ta'minlanadi (forward biased). Ushbu zanjir ikki dioddagi tokni taxminan o'zgarmas (approximately constant) ushlab turadi.



1-rasm: Differensial termometrik usul uchun elektr zanjiri.

Agar D_1 diodining harorati T_1 va D_2 diodining harorati T_2 bo'lsa, (1) ga asosan:

$$V_1(T_1) = V_1(T_0) - \alpha(T_1 - T_0)$$

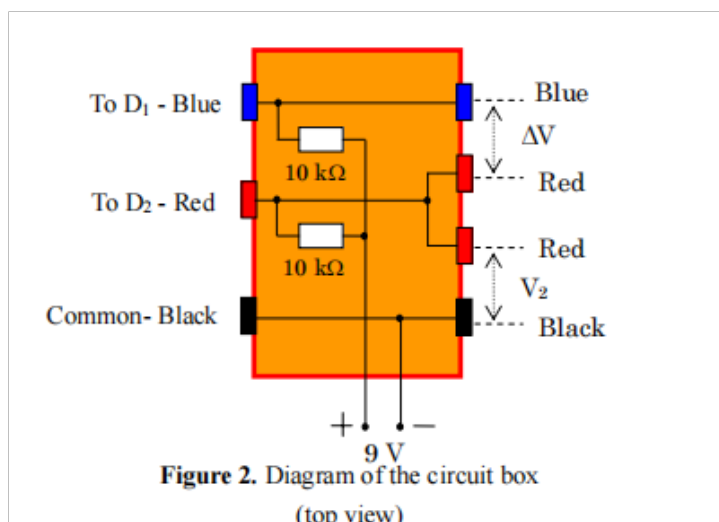
$$V_2(T_2) = V_2(T_0) - \alpha(T_2 - T_0)$$

Differensial kuchlanish (differential voltage):

$$\Delta V = V_2(T_2) - V_1(T_1) = V_2(T_0) - V_1(T_0) - \alpha(T_2 - T_1) = \Delta V(T_0) - \alpha\Delta T$$

bu yerda $\Delta T = T_2 - T_1$. Differensial kuchlanish ΔV ni o'lchab, harorat farqini aniqlash mumkin.

Diodlarni kuchlanish bilan ta'minlash uchun (to bias the diodes) sxema qutisi (circuit box) dan foydalanamiz, uning diagrammasi 2-rasmda (Rasm (2).png) ko'rsatilgan.



2-rasm: Sxema qutisining diagrammasi (ustki ko'rinish).

Sxema qutisida diodlar uchun ikkita 10 k kuchlanish beruvchi rezistor (biasing resistors), 9V batareyaga elektr simlari (electrical leads), D_1 va D_2 diodlariga ulanish uchun rozetkalar (sockets), va D_2 diodidagi kuchlanish pasayishi V_2 hamda D_1 va D_2 diodlarining differensial kuchlanishi ΔV ni o'lchash uchun raqamli multimetrlarga (digital multimeters) ulanish rozetkalari mavjud.

B. 1-topshiriq: Kristall moddaning qotish haroratini topish (Finding the temperature of solidification of a crystalline substance)

1. Tajriba maqsadi (Aim of the experiment)

Agar kristall qattiq modda eritilgan holatga (melting state) qadar qizdirilsa va keyin sovutilsa, u o'zgaras harorat T_s da qotadi (solidifies), bu harorat qotish harorati (temperature of solidification) yoki moddaning erish nuqtasi (melting point) deb ataladi. T_s ni aniqlashning an'anaviy usuli (traditional method) sovutish jarayonida haroratning vaqt bilan o'zgarishini kuzatishdir. Faza o'tishining yashirin issiqligi (latent heat) ajralib chiqishi tufayli, modda qotayotganda uning harorati o'zgarmaydi. Agar moddaning miqdori yetarlicha katta bo'lsa, harorat o'zgaras bo'lib turadigan vaqt oralig'i ancha uzoq bo'ladi va bu haroratni oson aniqlash mumkin. Aksincha, agar moddaning miqdori kichik bo'lsa, bu vaqt oralig'i kuzatish uchun juda qisqa va shuning uchun T_s ni aniqlash qiyin.

Kichik miqdordagi modda holatida T_s ni aniqlash uchun differensial termometrik usuldan foydalanamiz, uning printsipini quyidagicha umumlashtirish mumkin. Biz ikkita bir xil kichik idish (small dishes) dan foydalanamiz, biri o'rganilayotgan moddaning kichik miqdorini o'z ichiga oladi (namuna idishi – sample dish), ikkinchisi esa moddani o'z ichiga olmaydi (mos yozuvlar idishi – reference dish). Ikki idish vaqt bilan sekin o'zgaruvchan haroratli issiqlik manbaiga (heat source) qo'yiladi. Ikki idishga issiqlik oqimlari (thermal flows) deyarli bir xil. Har bir idishda harorat datchigi (oldinga yo'naltirilgan kremniy diodi) mavjud. Moddada faza o'zgarishi (phase change) bo'lmaganda, namuna idishining harorati T_{sample} va mos yozuvlar idishining harorati T_{ref} deyarli bir xil tezlikda o'zgaradi va shuning uchun $\Delta T = T_{\text{ref}} - T_{\text{sample}}$ bilan sekin o'zgaradi. Agar moddada faza o'zgarishi sodir bo'lsa va faza o'zgarishi davomida T_{sample} o'zgarmaydi va T_s ga teng bo'lsa, T_{ref} esa barqaror o'zgaradi, u holda ΔT tez o'zgaradi. ΔT ning T_{sample} ga nisbatan grafigi

keskin o'zgarishni ko'rsatadi. ΔT ning keskin o'zgarishiga mos keladigan T_{sample} qiymati haqiqatda T_s dir.

Ushbu tajribaning maqsadi an'anaviy va differensial termal tahlil usullaridan (traditional and differential thermal analysis methods) foydalanib, qotish harorati T_s 50°C dan 70°C gacha bo'lgan sof kristall moddaning qotish haroratini aniqlashdir. Tajribada ishlatiladigan moddaning miqdori taxminan 20 mg.

2. Asboblari va materiallar (Apparatus and materials)

1. Issiqlik manbai (heat source): 20 Vt halogen lampa (halogen lamp).
2. Idish ushlagichi (dish holder): kvadrat teshikli bakelit plastinka (bakelite plate). Teshik ustiga po'lat plastinka (steel plate) mahkamlangan. Po'lat plastinkaga ikkita kichik magnit (small magnets) qo'yilgan.
3. Ikkita kichik po'lat idish (small steel dishes), har biriga kremniy diod lehimlangan (soldered). Bir idish mos yozuvlar idishi (reference dish), ikkinchisi – namuna idishi (sample dish) sifatida ishlatiladi.

Har bir idish magnit ustiga qo'yiladi. Magnit kuchi idish, magnit va po'lat plastinka o'rtasidagi kontakti ushlab turadi. Magnitlar, shuningdek, po'lat plastinka va idishlar o'rtasida mo'tadil issiqlik kontaktini (moderate thermal contact) ta'minlaydi.

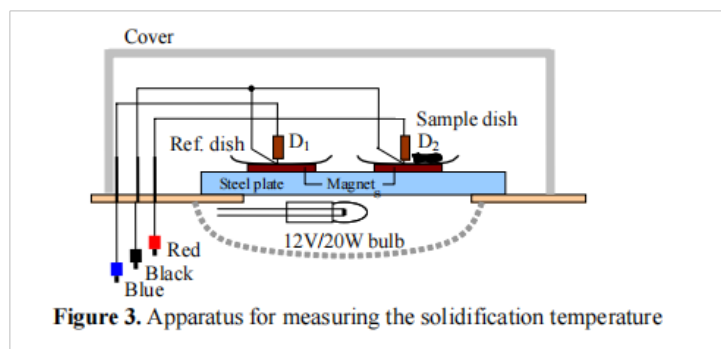
Kulrang plastik quti (grey plastic box) idishlarni tashqi ta'sirlardan himoya qilish uchun qopqoq (cover) sifatida ishlatiladi.

3-rasm (Rasm (3).png) da idish ushlagichida idishlar va magnitlarning joylashuvi va lampochka ko'rsatilgan.

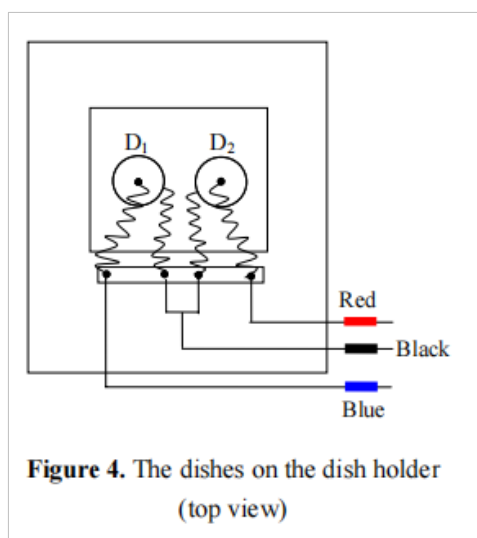
4. Ikkita raqamli multimetr (digital multimeters) voltmeter (voltmeter) sifatida ishlatiladi. Ular Funktsiya selektorini (Function selector) "°C/°F" funktsiyasiga aylantirib, xona haroratini ham o'lchay oladi. Multimetrning kuchlanish funktsiyasining xatoligi (error) oxirgi raqamda ± 2 ga teng.

Eslatma: Multimetr (9-rasmga qarang) "Auto power off" funktsiyasiga o'tib qolmasligi uchun, SELECT tugmasini bosib ushlab turganda, FUNC selektorini OFF holatidan kerakli funktsiyaga aylantiring.

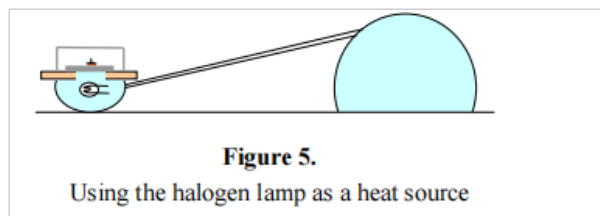
5. 2-rasmda ko'rsatilgan sxema qutisi (circuit box).
6. 9V batareya (9 V battery).
7. Elektr simlari (Electrical leads).
8. O'lchanadigan moddadan taxminan 20 mg bo'lgan kichik ampula (small ampoule).
9. Sekundomer (stop watch).
10. Kalkulyator (calculator).
11. Grafik qog'ozlari (Graph papers).



3-rasm: Idish ushlagichida idishlar va magnitlarning joylashishi (ustki ko'rinish).



4-rasm: Idish ushlagichidagi idishlar (ustki ko'rinish).



5-rasm: Halogen lampani issiqlik manbai sifatida ishlatish.

3. Tajriba (Experiment)

1. Magnitlar po'lat plastinkaning ikkita ekvivalent joyiga qo'yiladi. Mos yozuvlar idishi va bo'sh namuna idishi 4-rasmda ko'rsatilganidek magnitlar ustiga qo'yiladi. Biz chap tomondagi idishni mos yozuvlar idishi (reference dish) sifatida ishlatamiz, undagi diod D_1 (mos yozuvlar diodi – reference diode) va o'ng tomondagi idishni namuna idishi (sample dish) sifatida ishlatamiz, undagi diod D_2 (o'lchash diodi – measuring diode) deb ataladi.

Lampaning abajurini (lamp-shade) 5-rasmda ko'rsatilganidek teskari qilib qo'ying. Lampani yoqmang. Idish ushlagichini lampa ustiga qo'ying. Asboblarni shunday ulangki, siz D_2 diodidagi kuchlanish pasayishini, ya'ni $V_{\text{samp}} = V_2$ ni va differensial kuchlanish ΔV ni o'lchay olasiz.

Asboblarning va qurilmalarning isinish davri (warming up period) tufayli xatoliklarni bartaraf etish uchun, haqiqiy tajribalarni boshlashdan oldin to'liq o'lchash zanjirini taxminan 5 daqiqa davomida yoqilgan holda saqlash qat'iy tavsiya etiladi.

- 1.1. Xona harorati T_0 va namuna idishiga biriktirilgan D_2 diodidagi kuchlanish pasayishi $V_{\text{samp}}(T_0)$ ni xona haroratida T_0 o'lchang.
- 1.2. O'lchash diodidagi kuchlanish pasayishlarini $V_{\text{samp}}(50^\circ\text{C})$, $V_{\text{samp}}(70^\circ\text{C})$ va $V_{\text{samp}}(80^\circ\text{C})$ ni 50°C , 70°C va 80°C haroratlarda hisoblang.
2. Ikkala idish ham bo'sh holda, lampani yoqing. V_{samp} ni kuzating. Namuna idishining harorati $T_{\text{samp}} \sim 80^\circ\text{C}$ ga yetganda, lampani o'chiring.
- 2.1. $T_{\text{samp}} \sim 70^\circ\text{C}$ gacha kuting, so'ngra po'lat plastinka soviyotganda V_{samp} va ΔV ning vaqt bilan o'zgarishini kuzating. Har 10 s dan 20 s gacha V_{samp} va ΔV qiymatlarini javoblar varag'idagi jadvalga yozib boring. Agar ΔV tez o'zgarsa, ketma-ket o'lchovlar orasidagi vaqt oralig'i qisqaroq bo'lishi mumkin. Namuna idishining harorati $T_{\text{samp}} \sim 50^\circ\text{C}$ ga tushganda o'lchov to'xtatiladi.
- 2.2. Berilgan grafik qog'ozida V_{samp} ning t ga bog'liqlik grafigini (Graph 1 deb ataladi) chizing.
- 2.3. Berilgan grafik qog'ozida ΔV ning V_{samp} ga bog'liqlik grafigini (Graph 2 deb ataladi) chizing.
- Eslatma:** 2.2 va 2.3 uchun har bir grafikning to'g'ri nomini yozishni unutmang.
3. Ampuladagi moddani namuna idishiga to'kib soling (pour). 2-bo'limda aytilganidek tajribani xuddi shunday takrorlang.
- 3.1. V_{samp} va ΔV ning vaqt t ga bog'liq ma'lumotlarini javoblar varag'idagi jadvalga yozing.
- 3.2. V_{samp} ning t ga bog'liqlik grafigini (Graph 3) chizing.
- 3.3. ΔV ning V_{samp} ga bog'liqlik grafigini (Graph 4) chizing.
4. 2-bo'lim va 3-bo'limdagi grafiklarni solishtirib, moddaning qotish haroratini aniqlang.
- 4.1. An'anaviy usul (traditional method) yordamida T_s ni aniqlash: 3 va 2-bo'limlardagi V_{samp} ning t ga grafiklarini (ya'ni Graph 3 va Graph 1) solishtirib, modda qotadigan nuqtani Graph 3 da belgilang va V_{samp} ning shu nuqtaga mos keladigan qiymati V_s ni aniqlang. Moddaning qotish harorati T_s ni toping va uning xatoligini baholang.
- 4.2. Differensial termometrik usul (differential thermometric method) yordamida T_s ni aniqlash: 3 va 2-bo'limlardagi ΔV ning V_{samp} ga grafiklarini (ya'ni Graph 4 va Graph 2) solishtirib, modda qotadigan nuqtani Graph 4 da belgilang va V_{samp} ning shu nuqtaga mos keladigan qiymati V_s ni aniqlang. Moddaning qotish harorati T_s ni toping.
- 4.3. O'lchov ma'lumotlari va asboblarning xatoliklaridan (errors of measurement data and instruments) foydalanib, differensial termometrik usul bilan olingan T_s ning xatoligini hisoblang. Xatolik hisoblarini yozing va nihoyat T_s qiymatlarini xatoligi bilan birga javoblar varag'iga yozing.

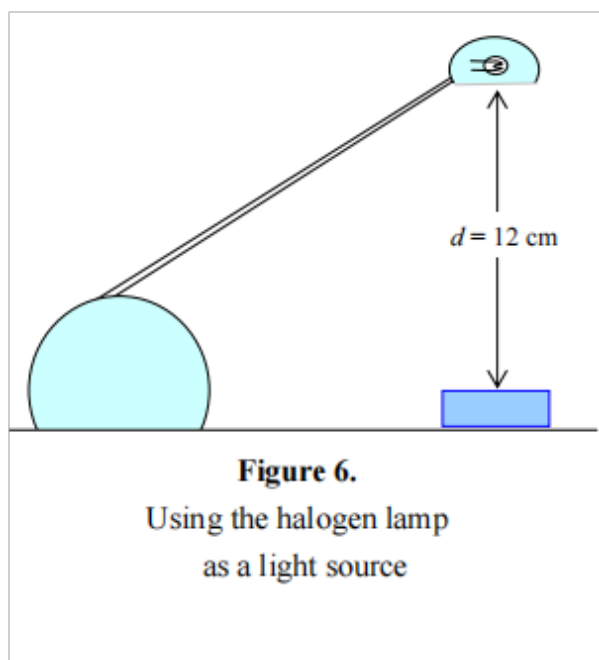
C. 2-topshiriq: Quyosh batareyasining samaradorligini cho'g'lanma lampa yoritgichida aniqlash (Determining the efficiency of a solar cell under illumination of an incandescent lamp)

1. Tajriba maqsadi (Aim of the experiment)

Tajribaning maqsadi cho'g'lanma lampa (incandescent lamp) yoritgichida quyosh batareyasining (solar cell) samaradorligini (efficiency) aniqlashdir. Samaradorlik – quyosh batareyasining tashqi zanjirga (external circuit) bera oladigan elektr quvvatining (electrical power) batareya olgan umumiy nurlanish quvvatiga (total radiant power) nisbati sifatida aniqlanadi. Samaradorlik tushayotgan nurlanish spektriga (incident radiation spectrum) bog'liq. Ushbu tajribada batareyaga tushadigan nurlanish – bu cho'g'lanma halogen lampanikidir. Quyosh batareyasining samaradorligini aniqlash uchun, biz lampa ostida, lampadan vertikal yo'nalishda d masofada joylashgan nuqtadagi nurlanish kuchlanganligini (irradiance) E va quyosh batareyasi shu nuqtaga qo'yilganda uning maksimal quvvatini $P_{\max}$ o'lchashimiz kerak. Ushbu tajribada $d = 12\text{ sm}$ (6-rasm – Rasm (6).png). Nurlanish kuchlanganligi E quyidagicha aniqlanadi:

$$E = \Phi/S$$

bu yerda Φ – nurlanish oqimi (radiant flux – nurlanish quvvati), S – yoritilgan sirtning (illuminated surface) maydoni.



6-rasm: Halogen lampani yorug'lik manbai sifatida ishlatish.

2. Asboblari va materiallari (Apparatus and materials)

1. Yorug'lik manbai (light source): 20 Vt halogen lampa.
2. Nurlanish detektori (radiation detector): misdan (copper) yasalgan ichi bo'sh konus (hollow cone), uning ichki yuzasi kuyik (soot) bilan qoraytirilgan (7-rasm – Rasm (7).png). Konus atrof-muhitdan to'liq bo'lmagan issiqlik izolyatsiyasiga ega (incompletely thermally isolated). Ushbu tajribada detektor ideal qora jism (ideal

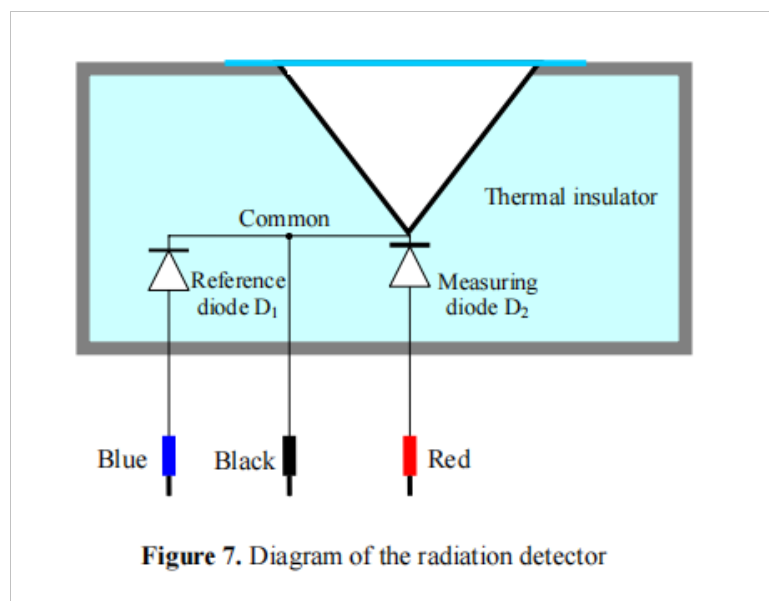
black body) deb hisoblanadi. Haroratni o'lchash uchun kremniy diodlardan foydalanamiz. O'lchash diodi (measuring diode) nurlanish detektoriga mahkamlangan (1-rasm va 7-rasmdagi D_2), shuning uchun uning harorati konusning haroratiga teng. Mos yozuvlar diodi (reference diode) detektorni o'z ichiga olgan qutining devorining ichki tomoniga joylashtirilgan; uning harorati atrof-muhit haroratiga teng. Detektorning (konus va o'lchash diodi) umumiy issiqlik sig'imi (total heat capacity) $C = (0.69 \pm 0.02) \text{ J/K}$. Detektor juda yupqa polietilen plyonka (polyethylene film) bilan qoplangan; uning nurlanish yutishi va qaytarishi (radiation absorption and reflection) ahamiyatsiz.

3. 2-rasmda ko'rsatilgan sxema qutisi (circuit box).
4. Plastik qutiga mahkamlangan bir bo'lak quyosh batareyasi (solar cell) (8-rasm – Rasm (8).png). Batareyaning maydoni (area) bir nechta metall ulash chiziqlarini (metal connection strips) o'z ichiga oladi. Samaradorlik hisobi uchun bu chiziqlar batareyaning bir qismi deb hisoblanadi.
5. Ikkita raqamli multimetr (digital multimeters). Kuchlanishni (voltage) o'lchash uchun ishlatilganda, ular juda katta ichki qarshilikka (very large internal resistance) ega, uni cheksiz katta deb hisoblash mumkin. Tokni (current) o'lchash uchun ishlatilganda, ularning ichki qarshiligini e'tiborsiz qoldirib bo'lmaydi. Multimetrning kuchlanish funktsiyasining xatoligi oxirgi raqamda ± 2 ga teng.

Multimetrlar xona haroratini ham o'lchay oladi.

Eslatma: Multimetr (9-rasmga qarang) “Auto power off” funktsiyasiga o'tib qolmasligi uchun, SELECT tugmasini bosib ushlab turganda, FUNC selektorini OFF holatidan kerakli funktsiyaga aylantiring.

6. 9V batareya (9 V battery).
7. O'zgaruvchan qarshilik (variable resistor – reostat).
8. Sekundomer (stop watch).
9. 1 mm bo'linmalarga ega chizg'ich (ruler with 1mm divisions).
10. Elektr simlari (Electrical leads).
11. Grafik qog'ozlari (Graph papers).



7-rasm: Nurlanish detektori (konus).

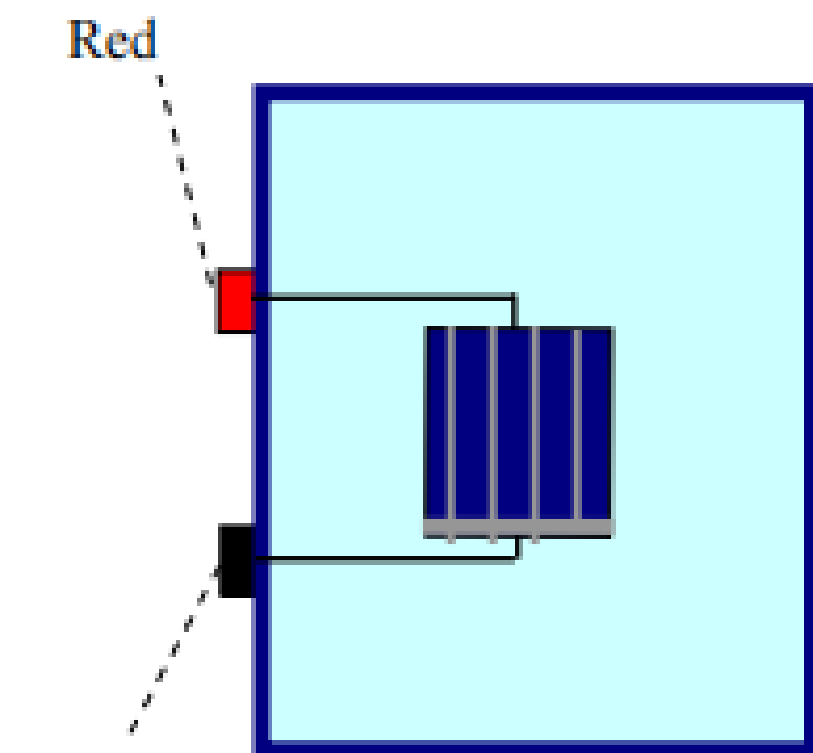
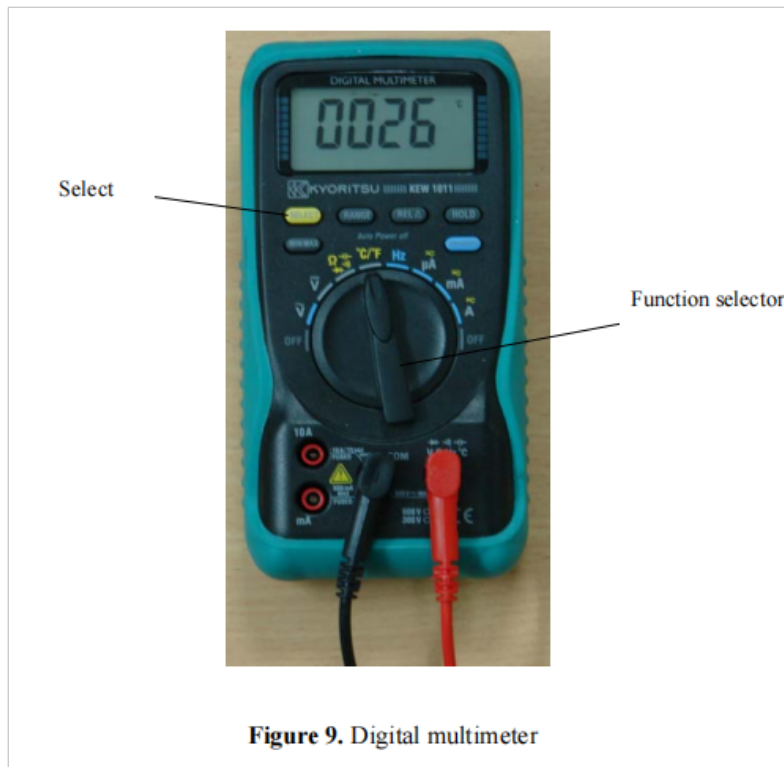


Figure 8.
The solar cell

8-rasm: Quyosh batareyasi plastik qutiga mahkamlangan.



9-rasm: Raqamli multimetr.

3. Tajriba (Experiment)

Detektor nurlanishdan energiya qabul qilganda, u qiziydi (heats up). Shu bilan birga, detektor issiqlikni bir necha mexanizmlar – issiqlik o'tkazuvchanlik (thermal conduction), konveksiya (convection), nurlanish (radiation) va hokazo – orqali yo'qotadi. Shunday qilib, detektor tomonidan dt vaqt oralig'ida olingan nurlanish energiyasi detektor haroratini oshirish uchun zarur energiya va detektordan atrofga o'tkazilgan energiyaning yig'indisiga teng:

$$\Phi dt = CdT + dQ$$

bu yerda C – detektor va diodning issiqlik sig'imi, dT – harorat o'sishi, dQ – issiqlik yo'qotilishi.

Detektor va atrof-muhit o'rtasidagi harorat farqi $\Delta T = T - T_0$ kichik bo'lganda, dt vaqt oralig'ida detektordan atrofga o'tkazilgan issiqlik dQ taxminan ΔT va dt ga proporsional deb hisoblash mumkin, ya'ni $dQ = k\Delta T dt$, bu yerda k – o'lchovi W/K bo'lgan koeffitsiyent. Shunday qilib, k o'zgarmas va ΔT kichik deb faraz qilib, quyidagini olamiz:

$$\Phi dt = CdT + k\Delta T dt = Cd(\Delta T) + k\Delta T dt$$

yoki

$$\frac{d(\Delta T)}{dt} + \frac{k}{C}\Delta T = \frac{\Phi}{C} \quad (4)$$

Ushbu differensial tenglamaning yechimi, detektor doimiy nurlanish bilan yoritila boshlagan paytdan boshlab ($t = 0$ da $\Delta T = 0$ deb hisoblab) harorat farqi ΔT ning vaqt t bilan o'zgarishini aniqlaydi:

$$\Delta T(t) = \frac{\Phi}{k} \left(1 - e^{-\frac{k}{C}t}\right) \quad (5)$$

Nurlanish o'chirilganda, yuqoridagi differensial tenglama quyidagi ko'rinishga keladi:

$$\frac{d(\Delta T)}{dt} + \frac{k}{C}\Delta T = 0 \quad (6)$$

va harorat farqi ΔT vaqt bilan quyidagi formula bo'yicha o'zgaradi:

$$\Delta T(t) = \Delta T(0)e^{-\frac{k}{C}t} \quad (7)$$

bu yerda $\Delta T(0) - t = 0$ (o'lchov boshlangan payt) dagi harorat farqi.

1. Xona haroratini T_0 aniqlang.
 2. Diod datchiklari (diode sensors), sxema qutisi (circuit box) va multimetrlardan iborat elektr zanjirini detektorning haroratini o'lchash uchun tuzing.
Asboblarni va qurilmalarning isinish davri tufayli xatoliklarni bartaraf etish uchun, haqiqiy tajribalarni boshlashdan oldin to'liq o'lchash zanjirini taxminan 5 daqiqa davomida yoqilgan holda saqlash qat'iy tavsiya etiladi.
 - 2.1. Detektorni yorug'lik manbai ostiga, lampadan $d = 12$ sm masofaga qo'ying. Lampa o'chirilgan. ΔV ning o'zgarishini taxminan 2 daqiqa davomida 10 s oraliqlar bilan kuzating va (3) tenglamadagi $\Delta V(T_0)$ qiymatini aniqlang.
 - 2.2. Detektorni yoritish uchun lampani yoqing. ΔV ning o'zgarishini kuzating. Har 10-15 s da ΔV qiymatini javoblar varag'idagi jadvalga yozib boring. (Eslatma: jadvalning x va y ustunlari keyinchalik 4-bo'limda ishlatiladi). 2 daqiqadan so'ng lampani o'chiring.
 - 2.3. Detektorni lampadan uzoqlashtiring. Shundan keyin ΔV ning o'zgarishini taxminan 2 daqiqa davomida kuzating. Har 10-15 s da ΔV qiymatini javoblar varag'idagi jadvalga yozib boring. (Eslatma: jadvalning x va y ustunlari keyinchalik 3-bo'limda ishlatiladi).
- Maslahat (Hints):** Detektor termal inertsiyaga (thermal inertia) ega bo'lganligi sababli, detektor yoritila boshlagan yoki yoritilish to'xtagan paytdan keyin darhol olingan ba'zi ma'lumotlardan foydalanmaslik tavsiya etiladi.
3. Lampani o'chirgandan so'ng (7) tenglama bajarilishini isbotlash uchun x va y o'zgaruvchilari mos ravishda tanlangan x - y koordinata tizimida grafik chizing.
 - 3.1. x va y o'zgaruvchilari uchun ifodalarni yozing.
 - 3.2. y ning x ga bog'liqlik grafigini (Graph 5) chizing.
 - 3.3. Grafikdan k qiymatini aniqlang.
 4. Detektor yoritilganda (5) tenglama bajarilishini isbotlash uchun x va y o'zgaruvchilari mos ravishda tanlangan x - y koordinata tizimida grafik chizing.
 - 4.1. x va y o'zgaruvchilari uchun ifodalarni yozing.

- 4.2. y ning x ga bog'liqlik grafigini (Graph 6) chizing.
- 4.3. Detektor teshigidagi (orifice) nurlanish kuchlanganligini E aniqlang.
5. Quyosh batareyasini nurlanish detektori bo'lgan joyga qo'ying. Quyosh batareyasini multimetrlar va batareyaning yukini (load) o'zgartirish uchun ishlatiladigan o'zgaruvchan qarshilik (variable resistor) dan iborat mos elektr zanjiriga ulang. Qarshilikning turli qiymatlarida zanjirdagi tokni (current) va batareyadagi kuchlanishni (voltage) o'lchang.
- 5.1. Ushbu tajribada ishlatilgan zanjirning diagrammasini chizing.
- 5.2. O'zgaruvchan qarshilikning tugmachasini aylantirib, siz yukning qiymatini o'zgartirasiz. Tugmachaning har bir holatida tok I va kuchlanish V qiymatlarini yozib oling.
- 5.3. Batareyaning yukga beradigan quvvatini (power) batareyadan o'tadigan tokning funktsiyasi sifatida grafigini chizing. Bu Graph 7.
- 5.4. Grafikdan batareyaning maksimal quvvatini $P_{\max}$ aniqlang va uning xatoligini baholang.
- 5.5. Olingan maksimal quvvatga mos keladigan batareyaning samaradorligi (efficiency) uchun ifodani yozing. Uning qiymatini va xatoligini hisoblang.

Tajriba to'plamining mazmuni (Contents of the experiment kit)

(10-rasm – Rasm (10).png ga qarang)

#	Komponent (Component)	Miqdor
1	Halogen lampa 220 V/20 W	1
2	Idish ushlagichi (Dish holder)	1
3	Idish (Dish)	2
4	Multimetr (Multimeter)	2
5	Sxema qutisi (Circuit box)	1
6	9 V batareya (9 V battery)	1
7	Elektr simlari (Electrical leads)	1 to'plam
8	O'lchanadigan modda solingan ampula (Ampoule with substance to be measured)	1
9	Sekundomer (Stop watch)	1
10	Kalkulyator (Calculator)	1
11	Nurlanish detektori (Radiation detector)	1
12	Quyosh batareyasi (Solar cell)	1
13	O'zgaruvchan qarshilik (Variable resistor)	1
14	Chizg'ich (Ruler)	1
15	Qopqoq sifatida ishlatiladigan quti (Box used as a cover)	1

Eslatma: Multimetr (9-rasmga qarang) “Auto power off” funktsiyasiga o'tib qolmasligi uchun, SELECT tugmasini bosib ushlab turganda, FUNC selektorini OFF holatidan kerakli funktsiyaga aylantiring.

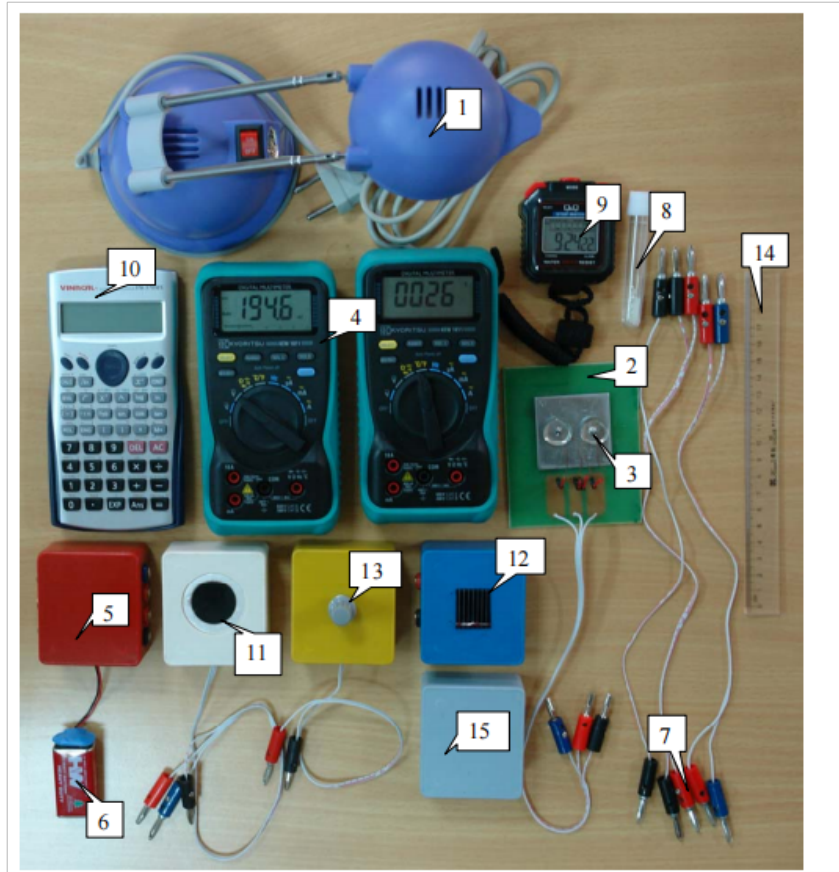


Figure 10. Contents of the experiment kit

10-rasm: Tajriba to'plamining mazmuni.

Yechim (Solution)

1. Kelining tuzilishi (The structure of the mortar)

1.1. TG masofasini hisoblash (Calculating the distance TG)

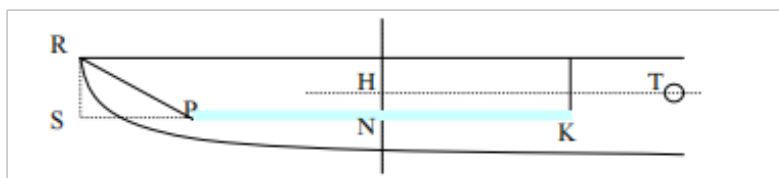
Chelakdagi suv hajmi (volume of water) $V = 1000 \text{ cm}^3 = 10^{-3} \text{ m}^3$. Chelak tubining uzunligi $d = L - h \tan 60^\circ = (0.74 - 0.12 \tan 60^\circ) \text{ m} = 0.5322 \text{ m}$.

(Boshlang'ich ma'lumotlar ikkita muhim raqam bilan berilganligi sababli, yakuniy javobda faqat ikkita muhim raqamni saqlaymiz, ammo oraliq bosqichlarda ko'proq raqamlarni saqlaymiz). Chelakdagi suv qatlamining balandligi c quyidagi formula bo'yicha hisoblanadi:

$$V = bcd + b \frac{c}{2} \tan 60^\circ \Rightarrow c = \frac{(d^2 + 2\sqrt{3}V/b)^{1/2} - d}{\sqrt{3}}$$

V , b va d uchun son qiymatlarni qo'yib, $c = 0.01228 \text{ m}$ topamiz.

Richag gorizontal yotganda, aylanish o'qi va suvning massa markazi (center of mass) N orasidagi gorizontal o'qdagi masofa $TH \approx a + \frac{d}{2} + \frac{c}{4} \tan 60^\circ = 0.4714 \text{ m}$ va $TG = (m/M)TH = 0.01571 \text{ m}$ (quyidagi rasmga qarang).



Kesim geometriyasi.

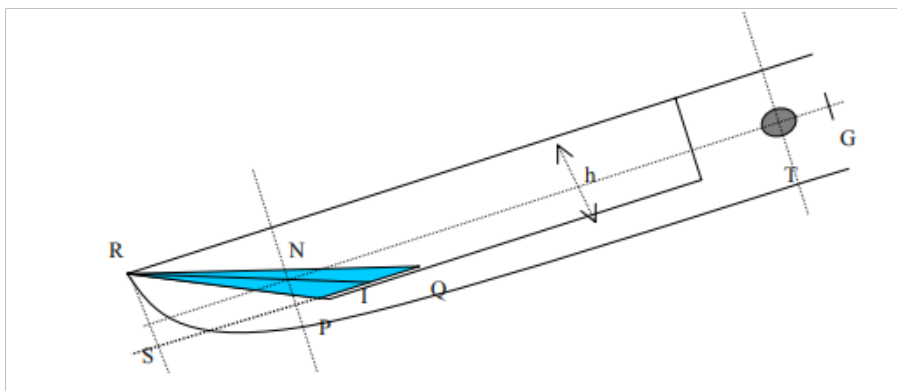
Javob: $TG = 0.016$ m.

1.2. α_1 va α_2 qiymatlarini hisoblash (Calculating the values of α_1 and α_2)

Richag α_1 burchakka og'ganida, suv sathi chelakning chetida bo'ladi. Bu nuqtada suv hajmi 10^{-3} m^3 ga teng. $PQ < d$ deb faraz qilaylik. Geometriyadan $V = hb \times PQ/2$, bundan $PQ = 0.1111$ m. $PQ < d$ farazi aniq bajariladi ($d = 0.5322$ m).

α_1 burchakni hisoblash uchun, $\tan \alpha_1 = h/QS = h/(PQ + \sqrt{3}h)$ ekanligini eslaymiz. Bundan $\alpha_1 = 20.6^\circ$ topiladi.

Og'ish burchagi 30° ga teng bo'lganda, chelak bo'shaydi: $\alpha_2 = 30^\circ$.



1.3. Richagning β qiyalik burchagini va umumiy moment μ nolga teng bo'lgan paytdagi chelakdagi suv miqdorini aniqlash

$PQ = x$ (m) deb belgilaymiz. Chelakdagi suv miqdori:

$$m = \rho_{\text{suv}} \frac{xhb}{2} = 9x \text{ (kg)}$$

$\mu = 0$ bo'lganda, chelakdagi suvdan keladigan moment richag og'irligidan keladigan momentni muvozanatlaydi. Rasmdagi suvning kesimi PQR uchburchagidir. Suvning massa markazi N RI medianasining $2/3$ qismida joylashgan, shuning uchun N , T va G bir to'g'ri chiziqli yotadi. U holda:

$$mg \times TN = Mg \times TG \quad \text{yoki} \quad m \times TN = M \times TG = 30 \times 0.01571 = 0.4714 \quad (1)$$

TN ni x orqali hisoblab, keyin (1) ga qo'yamiz:

$$TN = L + a - \frac{2}{3} \left(h\sqrt{3} + \frac{x}{2} \right) = 0.94 - 0.08\sqrt{3} - \frac{x}{3} = 0.8014 - \frac{x}{3}$$

Bu esa quyidagini anglatadi:

$$m \times TN = 9x \left(0.8014 - \frac{x}{3} \right) = -3x^2 + 7.213x \quad (2)$$

Shunday qilib, x uchun tenglama olamiz:

$$-3x^2 + 7.213x = 0.4714 \quad (3)$$

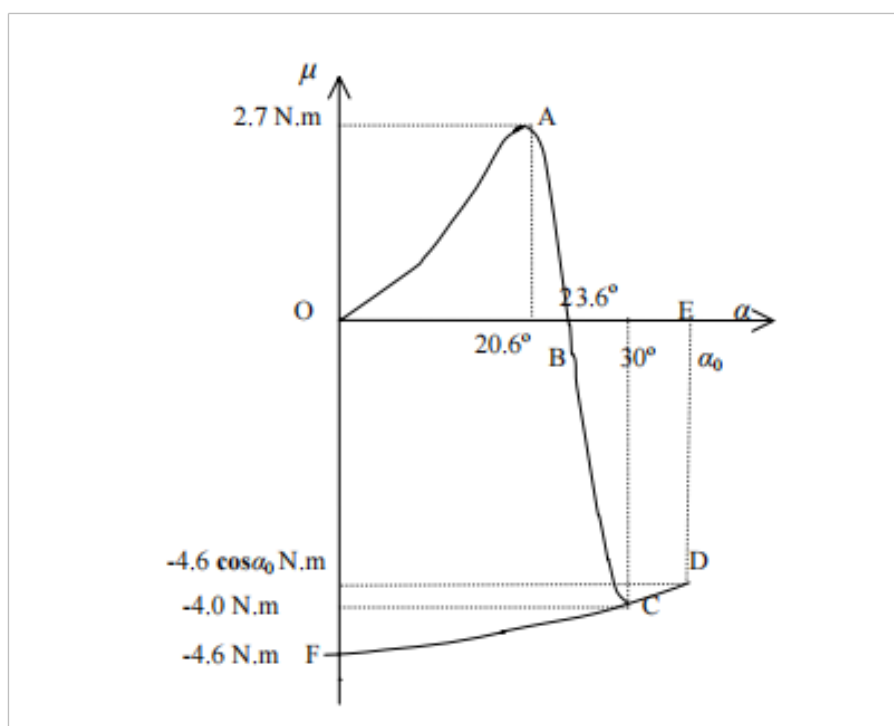
(3) tenglamaning yechimlari $x = 2.337$ va $x = 0.06723$. x 0.5322 dan kichik bo'lishi kerak, shuning uchun $x = x_0 = 0.06723$ va $m = 9x_0 = 0.6051$ kg ni olamiz.

$$\tan \beta = \frac{h}{x + h\sqrt{3}} = 0.4362, \quad \text{yoki} \quad \beta = 23.57^\circ$$

Javob: $m = 0.61$ kg va $\beta = 23.6^\circ$.

2. Ishchi rejim parametrlari (Parameters of the working mode)

2.1. Bir ishchi sikl davomida $\mu(\alpha)$, $\alpha(t)$ va $\mu(t)$ grafiklari



Bir ishchi siklda $\mu(\alpha)$ grafigi.

Dastlab chelakda suv yo'q, $\alpha = 0$, μ eng katta kattalikka ega, $gM \times TG = 30 \times 9.81 \times 0.01571 = 4.624$ N · m. Biz bu momentning ishorasini manfiy deb olamiz, chunki u α ni kamaytirishga intiladi.

Chelakka suv to'kilishi bilan suvdan keladigan moment (musbat ishora) μ ni oshiradi, to μ biroz musbat bo'lganda, richag ko'tarila boshlaydi. Shu paytdan boshlab, farazga ko'ra, chelakdagi suv miqdori o'zgarmas.

Richag qiyalashadi, shuning uchun suvning massa markazi aylanish o'qidan uzoqlashadi, bu esa μ ning oshishiga olib keladi. μ maksimalga suv chelakning chetidan to'kilish arafasida bo'lgan paytda erishadi. Bu paytda $\alpha = \alpha_1 = 20.6^\circ$.

Oddiy hisob-kitob:

$$SI = SP + PQ/2 = 0.12 \times 1.732 + 0.1111/2 = 0.2634 \text{ m}$$

$$TN = 0.20 + 0.74 - \frac{2}{3}SI = 0.7644 \text{ m}$$

$$\mu_{\max} = (1.0 \times TN - 30 \times TG)g \cos 20.6^\circ = (1.0 \times 0.7644 - 30 \times 0.01571) \times 9.81 \times \cos 20.6^\circ = 2.690 \text{ N} \cdot \text{m}$$

Demak, $\mu_{\max} = 2.7 \text{ N} \cdot \text{m}$.

Chelak yanada og'gan sayin, chelakdagi suv miqdori kamayadi va $\alpha = \beta$ bo'lganda, $\mu = 0$ ga teng bo'ladi. Inertsia tufayli α o'sishda davom etadi va μ kamayishda davom etadi. Chelak $\alpha = 30^\circ$ da bo'shaydi, bu paytda μ quyidagiga teng:

$$-30 \times g \times TG \times \cos 30^\circ = -4.0 \text{ N} \cdot \text{m}$$

Undan keyin inertsia tufayli α α_0 gacha ($\mu = -gMTG \cos \alpha_0 = -4.62 \cos \alpha_0 \text{ N} \cdot \text{m}$) o'sadi, so'ngra tezda 0 ga ($\mu = -4.62 \text{ N} \cdot \text{m}$) kamayadi.

Shu asosda biz $\alpha(t)$, $\mu(t)$ va $\mu(\alpha)$ grafiklarini chizamiz.

2.2.

$\mu(\alpha)$ moment tomonidan bajarilgan cheksiz kichik ish $dW = \mu(\alpha)d\alpha$ ga teng. Bir sikl davomida richagning $\mu(\alpha)$ ta'sirida olgan energiyasi $W = \oint \mu(\alpha)d\alpha$ bo'lib, bu $\mu(\alpha)$ chizig'i bilan chegaralangan yuza (area) dir. Shuning uchun W_{total} $\mu(\alpha)$ grafigidagi (OABCDFO) egri chizig'i bilan chegaralangan yuzaga teng.

Richag keliga (mortar) o'tkazadigan ish (work that the lever transfers to the mortar) – bu richag $\alpha = \alpha_0$ holatidan gorizontol holatga ($\alpha = 0$) o'tganda olgan energiyadir. Bizda W_{pounding} $\mu(\alpha)$ grafigidagi (OEDFO) yuzaga teng. U $gM \times TG \times \sin \alpha_0 = 4.6 \sin \alpha_0$ (J) ga teng.

2.3.

α_0 kattaligini D nuqtada richag energiyasi nolga teng ekanligidan baholash mumkin:

$$\text{yuza (OABO)} = \text{yuza (BEDCB)}$$

OABO ni uchburchak, BEDCB ni trapetsiya deb yaqinlashtirib:

$$23.6 \times 2.7 \times \frac{1}{2} = 4.0 \times [(\alpha_0 - 23.6) + (\alpha_0 - 30)] \times \frac{1}{2}$$

Bu esa $\alpha_0 = 34.7^\circ$ ni beradi. Bundan:

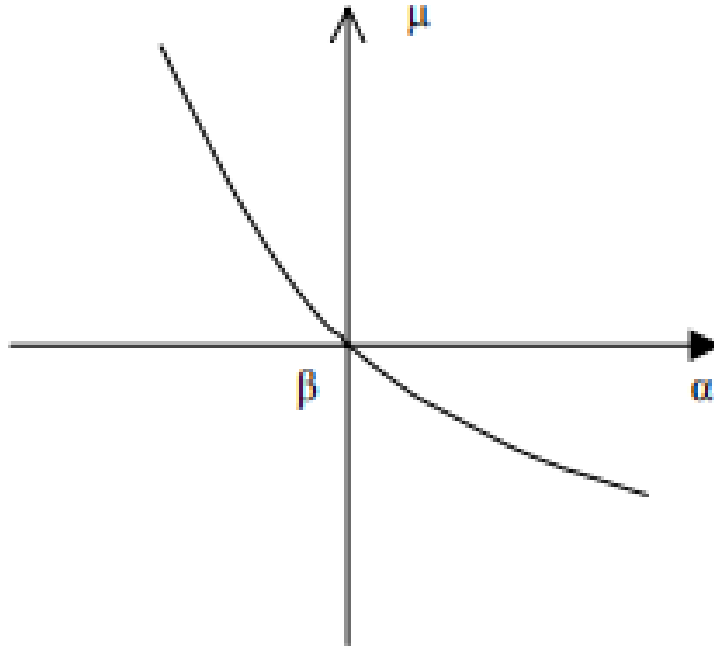
$$W_{\text{pounding}} = \text{yuza (OEDFO)} = \int_{34.76}^0 -Mg \times TG \times \cos \alpha d\alpha = 4.62 \times \sin 34.7^\circ = 2.63 \text{ J}$$

Shunday qilib, $W_{\text{pounding}} \approx 2.6 \text{ J}$.

3. Tinch rejim (The rest mode)

3.1. Chelak doimo to'lib toshib turadi (The bucket is always overflowing with water).

$\alpha = \beta$ atrofida $\mu(\alpha)$ ning ortib va kamayib boruvchi tarmoqlari bir-biriga mos keladi.



Grafik $\alpha = \beta$ kelining barqaror muvozanati (stable equilibrium) ekanligini ko'rsatadi.

3.1.2. $\alpha = \beta + \Delta\alpha$ ($\Delta\alpha$ kichik) bo'lganda μ moment uchun ifodani toping.

Richag α burchakka og'ganida chelakdagi suv massasi:

$$m = \frac{1}{2}\rho b h PQ, \quad \text{bu yerda } PQ = h \left(\frac{1}{\tan \alpha} - \frac{1}{\tan 30^\circ} \right)$$

Oddiy hisob-kitob shuni ko'rsatadiki, α β dan $\beta + \Delta\alpha$ ga oshganda, suv massasi quyidagicha o'zgaradi:

$$\Delta m = -\frac{bh^2\rho}{2\sin^2\alpha}\Delta\alpha \approx -\frac{bh^2\rho}{2\sin^2\beta}\Delta\alpha$$

Richagga ta'sir qiluvchi moment μ , og'ish $\beta + \Delta\alpha$ bo'lganda, Δm tufayli yuzaga kelgan momentga teng:

$$\mu = \Delta m \times g \times TN \times \cos(\beta + \Delta\alpha)$$

TN richagning β qiyalik burchagidagi muvozanat shartidan topiladi:

$$TN = M \times TG/m = 30 \times 0.01571/0.605 = 0.779 \text{ m}$$

Yakunida quyidagini olamiz:

$$\mu = -47.2 \times \Delta\alpha \text{ N} \cdot \text{m} \approx -47 \times \Delta\alpha \text{ N} \cdot \text{m}$$

3.1.3. Richagning harakat tenglamasi (Equation of motion of the lever)

$$\mu = I \frac{d^2\alpha}{dt^2}$$

bu yerda $\mu = -47 \times \Delta\alpha$, $\alpha = \beta + \Delta\alpha$, va I – richag va chelakdagi suvning T o'qiga nisbatan inersiya momentlarining yig'indisidir. Bu yerda I o'zgarmas emas, chunki chelakdagi suv miqdori α ga bog'liq. $\Delta\alpha$ kichik bo'lganda, suv miqdori va shaklini o'zgarmas deb hisoblash mumkin, shuning uchun I taxminan o'zgarmas. Chelakdagi suvni massasi 0.6 kg bo'lgan moddiy nuqta deb hisoblab, oddiy hisob:

$$I = 12 + 0.6 \times 0.78^2 = 12.36 \approx 12.4 \text{ kg} \cdot \text{m}^2$$

U holda:

$$-47 \times \Delta\alpha = 12.4 \times \frac{d^2 \Delta\alpha}{dt^2}$$

Bu garmonik ossillyator tenglamasidir. Tebranish davri (period):

$$\tau = 2\pi \sqrt{\frac{12.4}{47}} = 3.227 \text{ s}$$

Javob: $\tau = 3.2 \text{ s}$.

3.2. Chelak doimo to'lib toshgan holda richagning garmonik tebranishi ($\alpha = \beta$ atrofida)

Richag $\alpha = \beta$ atrofida amplitudasi $\Delta\alpha_0$ bilan garmonik tebranayotgan deb faraz qilaylik. $t = 0$ da $\Delta\alpha = 0$, chelak to'lib toshgan. dt vaqtda qiyalik $d\alpha$ ga o'zgaradi. Bizni $d\alpha < 0$ holati qiziqtiradi, ya'ni richag α kamayadigan tomonga harakat qilmoqda va chelakni to'lib toshishi uchun qo'shimcha suv kerak. Harakat tenglamasi:

$$\Delta\alpha = -\Delta\alpha_0 \sin\left(\frac{2\pi t}{\tau}\right), \quad \text{shuning uchun} \quad d(\Delta\alpha) = d\alpha = -\Delta\alpha_0 \frac{2\pi}{\tau} \cos\left(\frac{2\pi t}{\tau}\right) dt.$$

Chelak to'lib toshishi uchun, shu vaqt ichida chelakka tushayotgan suv miqdori kamida quyidagicha bo'lishi kerak:

$$dm = -\frac{bh^2\rho}{2\sin^2\beta} d\alpha = \frac{\Delta\alpha_0\tau bh^2\rho}{2\tau\sin^2\beta} \cos\left(\frac{2\pi t}{\tau}\right) dt.$$

dm maksimal $t = 0$ da,

$$dm_0 = \frac{\tau bh^2\rho\Delta\alpha_0}{\tau\sin^2\beta} dt.$$

Chelakka tushadigan suv miqdori oqim tezligi Φ bilan bog'liq: $dm_0 = \Phi dt$, shuning uchun

$$\Phi = \frac{\tau bh^2\rho\Delta\alpha_0}{\tau\sin^2\beta}.$$

To'lib toshgan chelak richagning garmonik tebranishlari uchun zarur shartdir, shuning uchun richagning amplitudasi $1^\circ = 2\pi/360$ rad bo'lgan garmonik tebranishlarga ega bo'lishi uchun $\Phi \geq \Phi_1$ bo'lishi kerak, bunda

$$\Phi_1 = \frac{\tau bh^2\rho \cdot 2\pi}{360\tau\sin^2\beta} = 0.2309 \text{ kg/s}$$

Demak, $\Phi_1 = 0.23 \text{ kg/s}$.

3.3. Φ_2 ni aniqlash

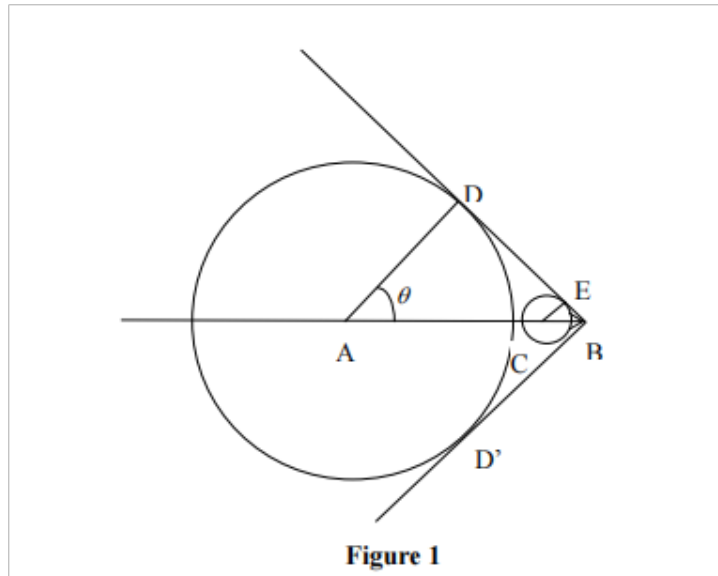
Agar qiyalik 20.6° gacha kamayganda chelak to'lib toshgan bo'lsa, u holda bu vaqtda chelakdagi suv miqdori 1 kg ga yetishi kerak va richag amplitudasi $23.6^\circ - 20.6^\circ = 3^\circ$ bo'lgan garmonik tebranadi. Oqim tezligi $3\Phi_1$ dan oshishi kerak, shuning uchun

$$\Phi_2 = 3 \times 0.23 \approx 0.7 \text{ kg/s.}$$

Bu sholi kelining ishlamasligi uchun minimal oqim tezligidir (minimal flow rate for the rice-pounding mortar not to work).

Cherenkov yorug'ligi va halqaviy tasvir hisoblagichi – Yechimlar

1.



1-rasm: Tolqin fronti va zarra traektoriyasi.

Zarra traektoriyasini oz ichiga olgan tekislikni korib chiqaylik. $t = 0$ da zarra A nuqtada. $t = t_1$ da B nuqtaga yetadi. Gyuygens prinsipiga kora, $0 < t < t_1$ vaqtda A da chiqarilgan nurlanish radiusi AD bolgan aylanaga, C da chiqarilgan nurlanish esa radiusi CE bolgan aylanaga yetadi. Sferalar radiuslari ularning markazining B gacha masofasiga proporsional:

$$\frac{CE}{CB} = \frac{c(t_1 - t)}{v(t_1 - t)} = \frac{1}{\beta n} = \text{const}$$

Shuning uchun sferalar B uchidan gomotetiya (homothety) bilan bir-biriga aylanadi va ularning konverti (envelope) B uchli va yarim ochilish burchagi $\varphi = \arcsin \frac{1}{\beta n} = \frac{\pi}{2} - \theta$ bolgan konusdir, bu yerda θ – yorug'lik nuri CE va zarra traektoriyasi orasidagi burchak.

1.1

Tolqin frontining tekislik bilan kesishmasi ikkita tog'ri chiziq, BD va BD' dir.

1.2

Ular zarra traektoriyasi bilan $\varphi = \arcsin \frac{1}{\beta n}$ burchak hosil qiladi.

2.

Zarralar dastasining halqa tasvirini topish uchun qurilma zarra traektoriyasi va kozguning optik oqini oz ichiga olgan tekislikda olinadi.

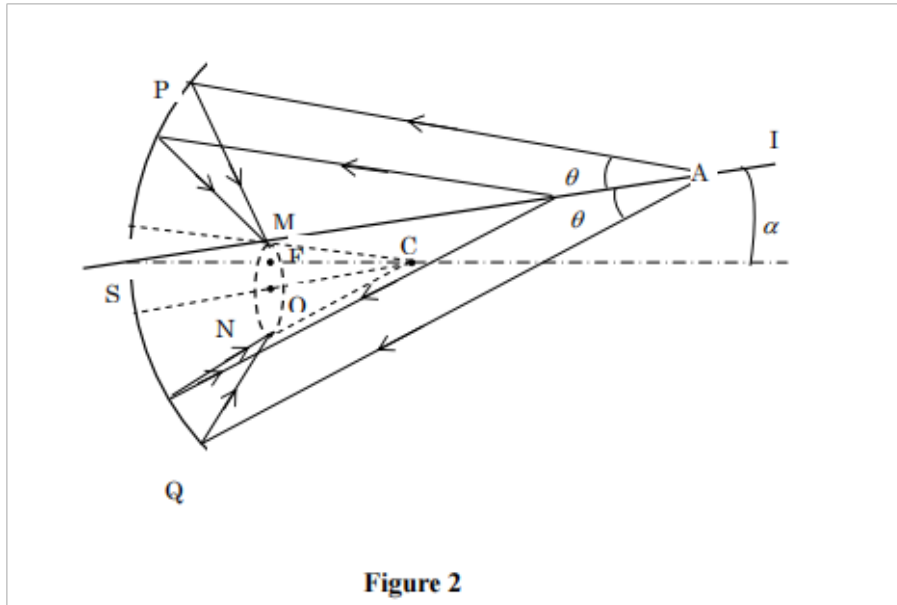


Figure 2
tasvirini hosil qilish geometriyasi.

2-rasm: Halqa

Belgilashlar:

- S – dasta sferik kozguga tushadigan nuqta.
- F – sferik kozguning fokusi.
- C – sferik kozguning markazi.
- IS – zaryadlangan zarra togri chiziqli traektoriyasi, kozguning optik oqi bilan kichik α burchak hosil qiladi.

$$CF = FS = f, CO \parallel IS, CM \parallel AP, CN \parallel AQ.$$

$$\angle FCO = \alpha \Rightarrow FO = f \cdot \alpha$$

$$\angle MCO = \angle OCN = \theta \Rightarrow MO = f \cdot \theta$$

C markazdan IS ga parallel togri chiziq otkazamiz. Bu chiziq fokal tekislikni O nuqtada kesadi. $FO \approx f \cdot \alpha$. C dan CO chizigining ikkala tomoniga u bilan θ burchak tashkil qiluvchi ikkita chiziq otkazamiz. Bu chiziqlar fokal tekislikni mos ravishda M va N nuqtalarda kesadi. Chizmada tekislikdagi barcha Cherenkov nurlari kozguga tushib, qaytgandan song M yoki N nuqtalarda kesishadi.

Uch olchamli holatda Cherenkov nurlanishi fokal tekislikda markazi O ($FO \approx f \cdot \alpha$) va radiusi $MO \approx f \cdot \theta$ bolgan halqa hosil qiladi.

3.

3.1

Cherenkov effekti sodir bolishi uchun $n > \frac{c}{v}$ yoki $n > \frac{1}{\beta}$ bolishi kerak. $n = 1 + \zeta$, $\zeta = 2.7 \cdot 10^{-4} P$ va $\frac{1}{\beta} = \sqrt{1 + K^2}$,

$$\zeta_{\min} = \sqrt{1 + K^2} - 1 \approx \frac{1}{2} K^2, \quad K = \frac{Mc}{p}$$

Shuning uchun

$$P_{\min} = \frac{1}{2.7 \cdot 10^{-4}} \cdot \frac{1}{2} K^2$$

Hisoblash natijalari:

- Protonlar: $P_{\min} = 16$ atm
- Kaonlar: $P_{\min} = 4.6$ atm
- Pionlar: $P_{\min} = 0.36$ atm

3.2

$\theta_\pi = 2\theta_\kappa$ shartidan foydalanib,

$$\cos \theta_\pi = \cos 2\theta_\kappa = 2 \cos^2 \theta_\kappa - 1$$

$\varepsilon = 1 - \beta \approx \frac{1}{2} K^2$ va $n = 1 + \zeta$ deb, yaqinlashishda:

$$\zeta_1 = \frac{4\varepsilon_\kappa - \varepsilon_\pi}{3} = \frac{1}{6} (4K_\kappa^2 - K_\pi^2)$$

$K_\kappa = 0.05$, $K_\pi = 0.014$ larni qoyib:

$$\zeta_1 = 1.62 \times 10^{-3}, \quad P_1 = \frac{\zeta_1}{2.7 \cdot 10^{-4}} = 6 \text{ atm}$$

$n = 1.00162$. $\theta_\kappa = 1.6^\circ$, $\theta_\pi = 3.2^\circ$. Protonlarning halqa tasviri kuzatilmaydi, chunki $P_1 = 6 \text{ atm} < 16 \text{ atm}$ (protonlar uchun minimal bosim).

4.

4.1

$\cos \theta = \frac{1}{\beta n}$ tenglamasidan logarifmik differensiallash:

$$\frac{\sin \theta \Delta \theta}{\cos \theta} = \frac{\Delta \beta}{\beta}$$

(3a) dan $\frac{\Delta \beta}{1 - \beta} = 2 \frac{\Delta p}{p}$. $1 - \beta \approx \frac{1}{2} K^2$, $\tan \theta \approx \theta$ ekanligidan:

$$\frac{\Delta \theta}{\Delta p} = \frac{2}{\theta} \cdot \frac{1 - \beta}{p\beta} = \frac{K^2}{\theta p}$$

Kaonlar: $K_\kappa = 0.05$, $\theta_\kappa = 1.6^\circ \Rightarrow \frac{\Delta \theta_\kappa}{\Delta p} = 0.51 \frac{1^\circ}{\text{GeV}/c}$. Pionlar: $K_\pi = 0.014$, $\theta_\pi = 3.2^\circ \Rightarrow \frac{\Delta \theta_\pi}{\Delta p} = 0.02 \frac{1^\circ}{\text{GeV}/c}$.

4.2

$$\frac{\Delta\theta_\kappa + \Delta\theta_\pi}{\Delta p} = 0.53 \frac{1^\circ}{\text{GeV}/c}$$

Ikki halqani ajratish sharti $\Delta\theta < 0.1(\theta_\pi - \theta_\kappa) = 0.16^\circ$. Bundan:

$$\Delta p < \frac{0.16}{0.53} = 0.3 \text{ GeV}/c$$

5.

5.1

Cherenkov effekti uchun β ning quyi chegarasi $\beta = \frac{1}{n} = \frac{1}{1.33}$. Zarraning kinetik energiyasi:

$$T = Mc^2 \left(\frac{1}{\sqrt{1 - \beta^2}} - 1 \right)$$

$\beta = 1/1.33$ ni qoyib:

$$T_{\min} = Mc^2 \left(\frac{1}{\sqrt{1 - (1/1.33)^2}} - 1 \right) = 0.517 Mc^2$$

5.2

α zarralari ($M_\alpha c^2 = 3.8 \text{ GeV}$): $T_{\min} = 0.517 \cdot 3.8 = 1.96 \text{ GeV}$. Elektronlar ($M_e c^2 = 0.51 \text{ MeV}$): $T_{\min} = 0.517 \cdot 0.51 = 0.264 \text{ MeV}$. Radioaktiv manbalar chiqaradigan zarralarning kinetik energiyasi bir necha MeV dan oshmaganligi sababli, Cherenkov nurlanishini elektronlar hosil qiladi.

6.

6.1

$P = 6 \text{ atm}$ da $\theta_\pi = 3.2^\circ$, $n = 1.00162$. Korinadigan diapazon chegaralarida ($\lambda = 0.4 \mu\text{m}$ va $0.8 \mu\text{m}$) sindirish koeffitsientlarining farqi $\delta n = 0.02(n - 1) = 3.24 \times 10^{-5}$. $\cos \theta = \frac{1}{\beta n}$ dan differensiallab:

$$\frac{\sin \theta \delta \theta}{\cos \theta} = \frac{\delta n}{n} \approx \delta n$$

$\tan \theta \approx \theta$ va $n \approx 1$ deb,

$$\delta \theta = \frac{\delta n}{\theta} = \frac{3.24 \times 10^{-5}}{3.2 \cdot \pi / 180} \approx 0.033^\circ$$

6.2

6.2.1 Dispersiya tufayli kengayish (yarim balandlikdagi yarim kenglik): $\frac{1}{2}\delta\theta = 0.017^\circ$.

6.2.2 Dastaning axromatikligi (momentumning tarqoqligi) tufayli kengayish: $0.02 \frac{1^\circ}{\text{GeV}/c} \times 0.3 \text{ GeV}/c = 0.006^\circ$.

6.2.3 Halqa rangi ichki chetidan tashqi chetiga qarab qizildan oqqa, songra kokka o'zgaradi.

Havo haroratining balandlik bilan ozgarishi, atmosfera barqarorligi va havo ifloslanishi – Yechimlar

1. Bosimning balandlik bilan o'zgarishi

Balandlik dz ga ozgarganda atmosfera bosimining ozgarishi:

$$dp = -\rho g dz \quad (1)$$

bu yerda g – erkin tushish tezlanishi (ozgarmas), ρ – havoning solishtirma massasi (specific mass). Havoni ideal gaz deb hisoblab:

$$\rho = \frac{m}{V} = \frac{p\mu}{RT}$$

(1) ga qoyamiz:

$$\frac{dp}{p} = -\frac{\mu g}{RT} dz$$

1.1

Agar havo harorati bir xil va T_0 ga teng bolsa:

$$\frac{dp}{p} = -\frac{\mu g}{RT_0} dz$$

Integrallash natijasida:

$$p(z) = p(0)e^{-\frac{\mu g}{RT_0}z} \quad (2)$$

1.2

Agar $T(z) = T(0) - \Lambda z$ bolsa (3):

$$\frac{dp}{p} = -\frac{\mu g}{R[T(0) - \Lambda z]} dz \quad (4)$$

1.2.1 $\int \frac{dz}{T(0) - \Lambda z} = -\frac{1}{\Lambda} \ln(T(0) - \Lambda z)$ ekanligidan foydalanib, (4) ni integrallaymiz:

$$\begin{aligned} \ln \frac{p(z)}{p(0)} &= \frac{\mu g}{R\Lambda} \ln \frac{T(0) - \Lambda z}{T(0)} = \frac{\mu g}{R\Lambda} \ln \left(1 - \frac{\Lambda z}{T(0)} \right) \\ p(z) &= p(0) \left(1 - \frac{\Lambda z}{T(0)} \right)^{\frac{\mu g}{R\Lambda}} \end{aligned} \quad (5)$$

1.2.2 Erkin konvektsiya (free convection) $\frac{\rho(z)}{\rho(0)} > 1$ bolganda sodir boladi. Zichliklar nisbati:

$$\frac{\rho(z)}{\rho(0)} = \frac{p(z) T(0)}{p(0) T(z)} = \left(1 - \frac{\Lambda z}{T(0)} \right)^{\frac{\mu g}{R\Lambda} - 1}$$

Bu ifoda 1 dan katta boladi, agar korsatkich manfiy bolsa:

$$\frac{\mu g}{R\Lambda} - 1 < 0 \quad \Rightarrow \quad \Lambda > \frac{\mu g}{R} = \frac{0.029 \cdot 9.81}{8.31} = 0.034 \text{ K/m}$$

2. Vertikal harakatdagi havo zarrasi haroratining o'zgarishi

Zarra bosimi har doim atrofdagi havo bosimiga teng. Zarra harorati T_{parcel} bosimga bogliq.

2.1

$$\frac{dT_{\text{parcel}}}{dz} = \frac{dT_{\text{parcel}}}{dp} \cdot \frac{dp}{dz}$$

Adiabatik jarayon uchun $pV^\gamma = \text{const}$ va holat tenglamasidan:

$$T_{\text{parcel}} p^{\frac{1-\gamma}{\gamma}} = \text{const} \quad (6)$$

Logarifmik differensiallab:

$$\begin{aligned} \frac{dT_{\text{parcel}}}{T_{\text{parcel}}} + \frac{1-\gamma}{\gamma} \frac{dp}{p} &= 0 \\ \Rightarrow \frac{dT_{\text{parcel}}}{dp} &= \frac{\gamma-1}{\gamma} \frac{T_{\text{parcel}}}{p} \end{aligned} \quad (7)$$

Bosim ozgarishi uchun (1) dan:

$$\frac{dp}{dz} = -\rho g = -\frac{p\mu g}{RT}$$

Shunday qilib:

$$\frac{dT_{\text{parcel}}}{dz} = -\frac{\gamma-1}{\gamma} \frac{\mu g}{R} \frac{T_{\text{parcel}}}{T} = -G \quad (8)$$

G odatda ozgarmas emas.

2.2

2.2.1 Agar istalgan balandlikda $T = T_{\text{parcel}}$ bolsa, (8) da G orniga:

$$\Gamma = \frac{\gamma-1}{\gamma} \frac{\mu g}{R} = \frac{\mu g}{c_p} \quad (9)$$

2.2.2 Son qiymati:

$$\Gamma = \frac{1.4-1}{1.4} \cdot \frac{0.029 \cdot 9.81}{8.31} \approx 0.00978 \text{ K/m} \approx 10^{-2} \text{ K/m}$$

2.2.3 Adiabatik atmosferada:

$$T(z) = T(0) - \Gamma z \quad (10)$$

2.3

$T(z) = T(0) - \Lambda z$ ni (7) ga qoyib:

$$\frac{dT_{\text{parcel}}}{T_{\text{parcel}}} = \frac{\gamma-1}{\gamma} \frac{\mu g}{R} \frac{dz}{T(0) - \Lambda z}$$

Integrallash:

$$\begin{aligned} \ln \frac{T_{\text{parcel}}(z)}{T_{\text{parcel}}(0)} &= \frac{\gamma-1}{\gamma} \frac{\mu g}{R} \cdot \frac{1}{\Lambda} \ln \frac{T(0) - \Lambda z}{T(0)} \\ T_{\text{parcel}}(z) &= T_{\text{parcel}}(0) \left(\frac{T(0) - \Lambda z}{T(0)} \right)^{\frac{\gamma}{\gamma-1}} \end{aligned} \quad (11)$$

2.4

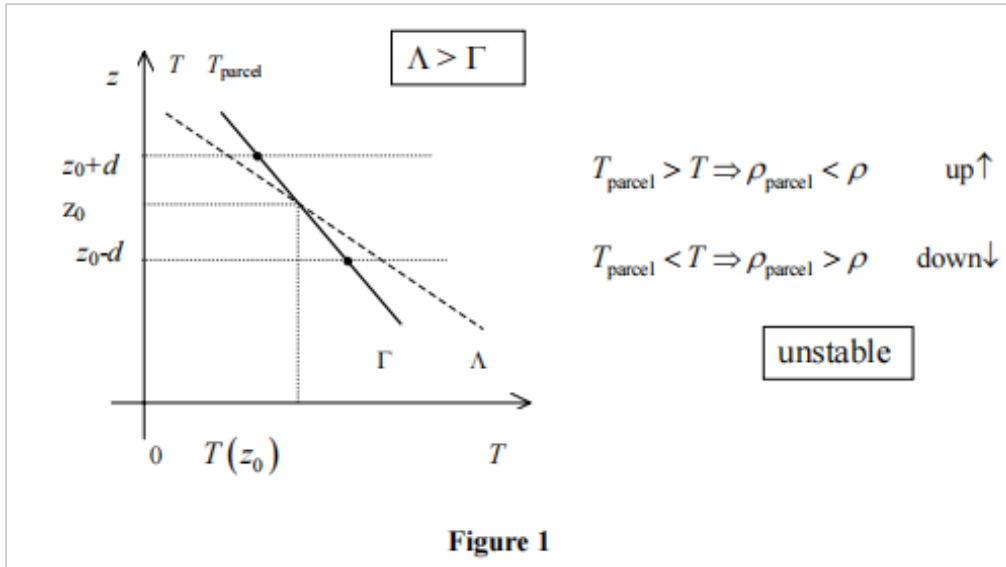
Agar $|\Lambda z| \ll T(0)$ va $T(0) \approx T_{\text{parcel}}(0)$ bolsa:

$$T_{\text{parcel}}(z) \approx T_{\text{parcel}}(0) - \Gamma z \quad (12)$$

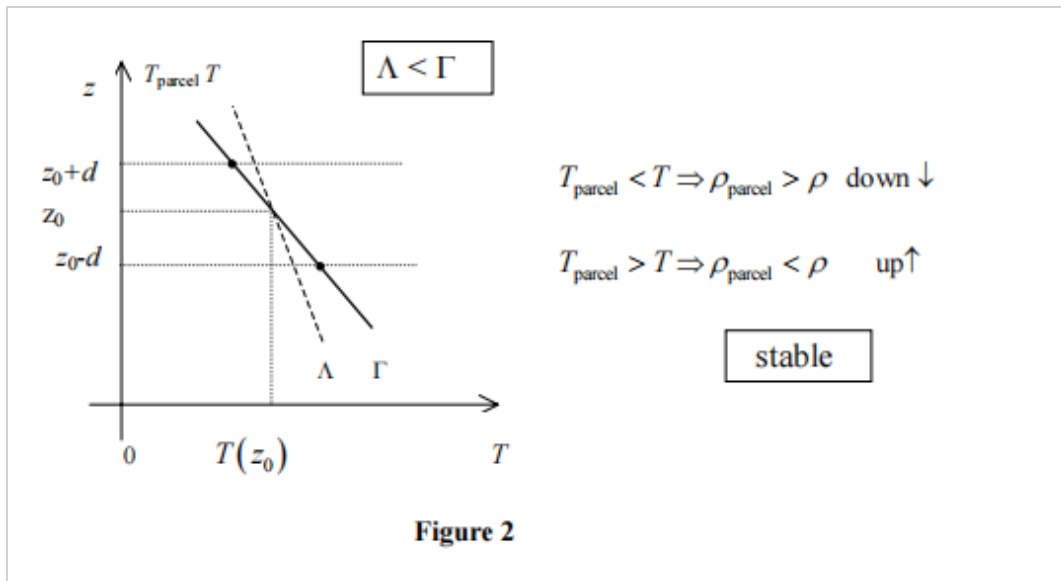
3. Atmosfera barqarorligi

z_0 balandlikda $T_{\text{parcel}}(z_0) = T(z_0)$ bolsa, zarra muvozanatda. $G \approx \Gamma$ deb hisoblab, barqarorlikni tekshiramiz.

- Agar $\Lambda > \Gamma$ (beqaror atmosfera): zarra yuqoriga kotarilsa $T_{\text{parcel}} > T$, zichlik kichikroq, itaruvchi kuch ogirlikdan katta – zarra uzoqlashadi.
- Agar $\Lambda < \Gamma$ (barqaror atmosfera): zarra yuqoriga kotarilsa $T_{\text{parcel}} < T$, zichlik katta, itaruvchi kuch kichik – zarra qaytadi.
- Agar $\Lambda = \Gamma$ (neytral): zarra har qanday holatda muvozanatda qoladi.



1-rasm: Beqaror atmosferada zarra harakati.



2-rasm: Barqaror atmosferada zarra harakati.

3.2

Barqaror atmosferada ($\Lambda < \Gamma$) yerdan $T_{\text{parcel}}(0) > T(0)$ bolgan zarra kotarilib, $T_{\text{parcel}}(h) = T(h)$ bolgan maksimal h balandlikka yetadi. (11) dan:

$$\left(1 - \frac{\Lambda h}{T(0)}\right)^{\frac{\Gamma}{\Lambda}} = \frac{T_{\text{parcel}}(0)}{T(h)} = \frac{T_{\text{parcel}}(0)}{T(0) \left(1 - \frac{\Lambda h}{T(0)}\right)}$$

Soddalashtirib:

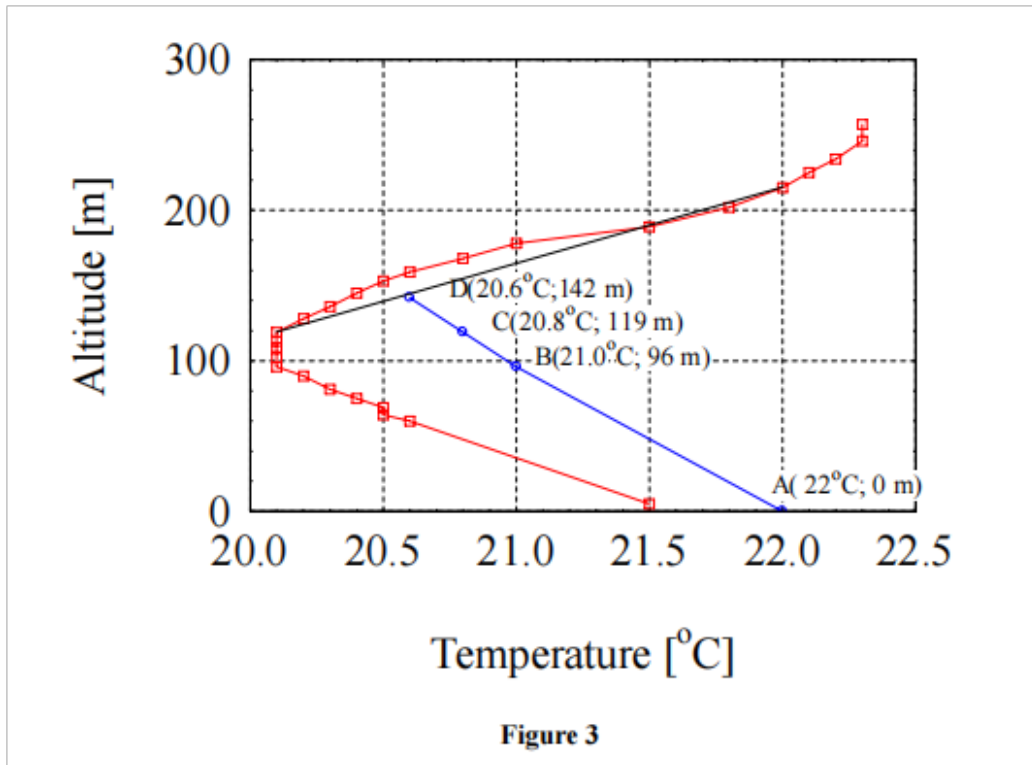
$$\left(1 - \frac{\Lambda h}{T(0)}\right)^{\frac{\Gamma}{\Lambda}-1} = \frac{T_{\text{parcel}}(0)}{T(0)}$$

Shuning uchun maksimal balandlik:

$$h = \frac{1}{\Lambda} \left[T(0) - \left(\frac{T(0)}{T_{\text{parcel}}(0)^{\Lambda}} \right)^{\frac{1}{\Gamma-\Lambda}} \right] \quad (13)$$

4. Aralashma balandligi (mixing height)

Jadval ma'lumotlari asosida z va T grafigi 3-rasmda keltirilgan.



3-rasm: Balandlik bo'yicha harorat.

4.1

Atmosferani 200 m dan pastda uch qatlamga ajratamiz:

1. $0 < z < 96$ m, $\Lambda_1 = \frac{21.5-20.1}{91} = 15.4 \times 10^{-3}$ K/m.
2. $96 < z < 119$ m, $\Lambda_2 = 0$ (izotermik qatlam).

3. $119 < z < 215$ m, $\Lambda_3 = -\frac{22-20.1}{215-119} = -0.02$ K/m.

1-qatlamda (11) dan:

$$T_{\text{parcel}}(96 \text{ m}) = 294.04 \text{ K} \approx 294.0 \text{ K} (21.0^\circ\text{C})$$

2-qatlamda izotermik atmosferadagi ifoda:

$$T_{\text{parcel}}(z) = T_{\text{parcel}}(0) \exp\left(-\frac{\Gamma z}{T(0)}\right)$$

Balandlik 96 m dan 119 m gacha ($\Delta z = 23$ m):

$$T_{\text{parcel}}(119 \text{ m}) = 293.81 \text{ K} (20.8^\circ\text{C})$$

4.2

3-qatlamda (13) dan $h = 23$ m, mos harorat $293.6 \text{ K} (20.6^\circ\text{C})$. Shunday qilib aralashma balandligi:

$$H = 119 + 23 = 142 \text{ m}$$

va $T_{\text{parcel}}(142 \text{ m}) = 293.6 \text{ K}$.

5. CO ifloslanishini baholash

CO chiqarish tezligi (7:00–8:00):

$$M = 800000 \times 5 \times 12/3600 = 13300 \text{ g/s}$$

To'rtburchak quti hajmi LWH , shamol tezligi u , CO konsentratsiyasi $C(t)$ (bir xil deb faraz qilinadi).

5.1

dt vaqt oralig'ida qutiga qo'shilgan CO massasi Mdt . Shamol olib chiqib ketadigan CO massasi $LHC(t)u dt$. O'zgarish qutidagi CO massasining o'zgarishiga teng:

$$Mdt - LHC(t)u dt = LWH dC$$

$$\frac{dC}{dt} + \frac{u}{W}C(t) = \frac{M}{LWH} \quad (14)$$

5.2

(14) ning umumiy yechimi:

$$C(t) = K \exp\left(-\frac{ut}{W}\right) + \frac{M}{LHu}$$

Boshlang'ich shart $C(0) = 0$ dan $K = -\frac{M}{LHu}$. Shuning uchun:

$$C(t) = \frac{M}{LHu} \left[1 - \exp\left(-\frac{ut}{W}\right)\right] \quad (16)$$

5.3

Ma'lumotlar: $L = 15 \text{ km} = 15000 \text{ m}$, $W = 8 \text{ km} = 8000 \text{ m}$, $u = 1 \text{ m/s}$, $t = 3600 \text{ s}$, $H = 142 \text{ m}$.

$$C(3600) = \frac{13300}{15000 \cdot 142 \cdot 1} \left[1 - \exp\left(-\frac{1 \cdot 3600}{8000}\right) \right] \\ = 6.25 \times 10^{-3} \cdot (1 - e^{-0.45}) \approx 6.25 \cdot 10^{-3} \cdot (1 - 0.6376) \approx 2.3 \text{ mg/m}^3$$

Differensial termometrik usul – Yechimlar (Differential Thermometric Method – Solution)

1-topshiriq (Task 1)

1.

1.1. $T_0 = 25 \pm 1^\circ\text{C}$

$$V_{\text{samp}}(T_0) = 573.9 \text{ mV}$$

Turli tajriba to'plamlarida V_{samp} yuqoridagi qiymatdan $\pm 40 \text{ mV}$ gacha farq qilishi mumkin.

Xatolik bahosi uchun eslatma (Note for error estimation):

δV va $\delta V'$ multimetrning xususiyatlaridan foydalanib hisoblanadi: ko'rsatkichning $\pm 0.5\%$ + oxirgi raqamda 2. Misol: agar $V = 500 \text{ mV}$ bo'lsa, xatolik $\delta V = 500 \times 0.5\% + 0.2 = 2.7 \text{ mV} \approx 3 \text{ mV}$.

Shunday qilib, $V_{\text{samp}}(T_0) = 574 \pm 3 \text{ mV}$.

$V_{\text{samp}}(T_0)$ ning 505 dan 585 mV gacha bo'lgan barcha qiymatlari qabul qilinadi.

1.2. Haroratni hisoblash formulasi (Formula for temperature calculation):

(1) tenglamadan: $V_{\text{samp}} = V_{\text{samp}}(T_0) - \alpha(T - T_0)$

$$V_{\text{samp}}(50^\circ\text{C}) = 523.9 \text{ mV}$$

$$V_{\text{samp}}(70^\circ\text{C}) = 483.9 \text{ mV}$$

$$V_{\text{samp}}(80^\circ\text{C}) = 463.9 \text{ mV}$$

Xatolik hisobi (Error calculation): $\delta V_{\text{samp}} = \delta V_{\text{samp}}(T_0) + (T - T_0)\delta\alpha$

Misol: $V_{\text{samp}} = 495.2 \text{ mV}$ bo'lsa, $\delta V_{\text{samp}} = 2.7 + 0.03 \times (50 - 25) = 3.45 \text{ mV} \approx 3.5 \text{ mV}$.

Shunday qilib:

$$V_{\text{samp}}(50^\circ\text{C}) = 524 \pm 4 \text{ mV}$$

$$V_{\text{samp}}(70^\circ\text{C}) = 484 \pm 4 \text{ mV}$$

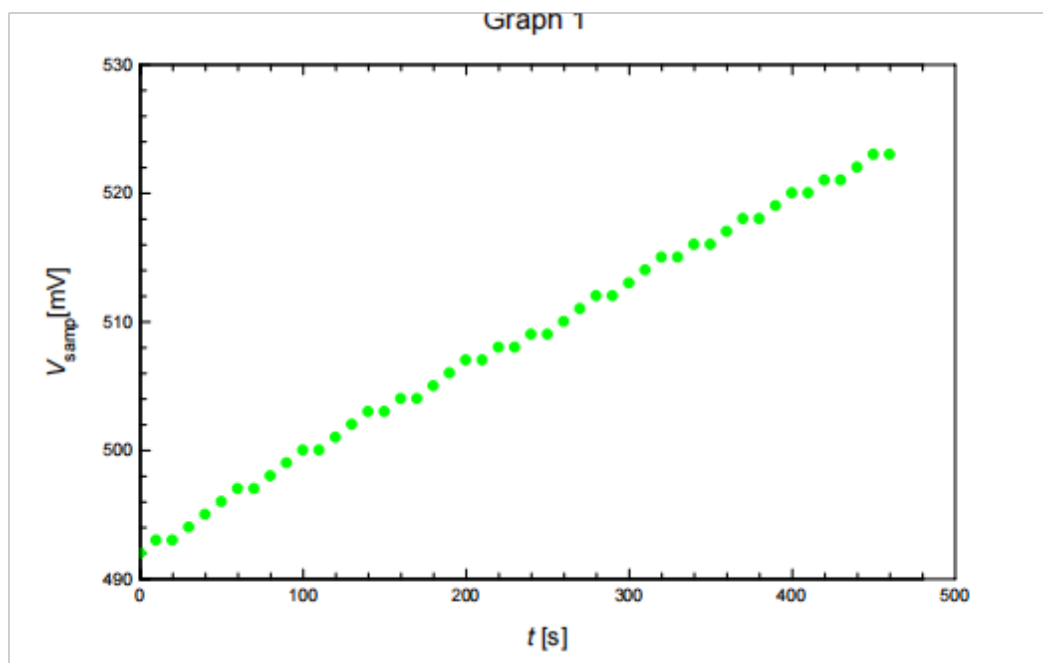
1.1 dagi kabi V_{samp} ning qabul qilinadigan oralig'i uchun xuddi shu qoida qo'llaniladi.

2. Namuna (modda) bo'lmagan holda sovutish jarayonining ma'lumotlari
(Data of cooling-down process without sample)

t (s)	V_{samp} (mV) (± 3 mV)	ΔV (mV) (± 0.2 mV)
0	492	-0.4
10	493	-0.5
20	493	-0.5
30	494	-0.6
40	495	-0.7
50	496	-0.7
60	497	-0.8
70	497	-0.8
80	498	-0.9
90	499	-1.0
100	500	-1.0
110	500	-1.1
120	501	-1.1
130	502	-1.2
140	503	-1.2
150	503	-1.3
160	504	-1.3
170	504	-1.4
180	505	-1.5
190	506	-1.6
200	507	-1.6
210	507	-1.7
220	508	-1.7
230	508	-1.8
240	509	-1.8
250	509	-1.8
260	510	-1.9
270	511	-1.9
280	512	-1.9
290	512	-2.0
300	513	-2.0
310	514	-2.1
320	515	-2.1
330	515	-2.1
340	516	-2.1
350	516	-2.2
360	517	-2.2
370	518	-2.3
380	518	-2.3
390	519	-2.3
400	520	-2.4
410	520	-2.4
420	521	-2.5
430	521	-2.5
440	522	-2.5
450	523	-2.6
460	523	-2.6

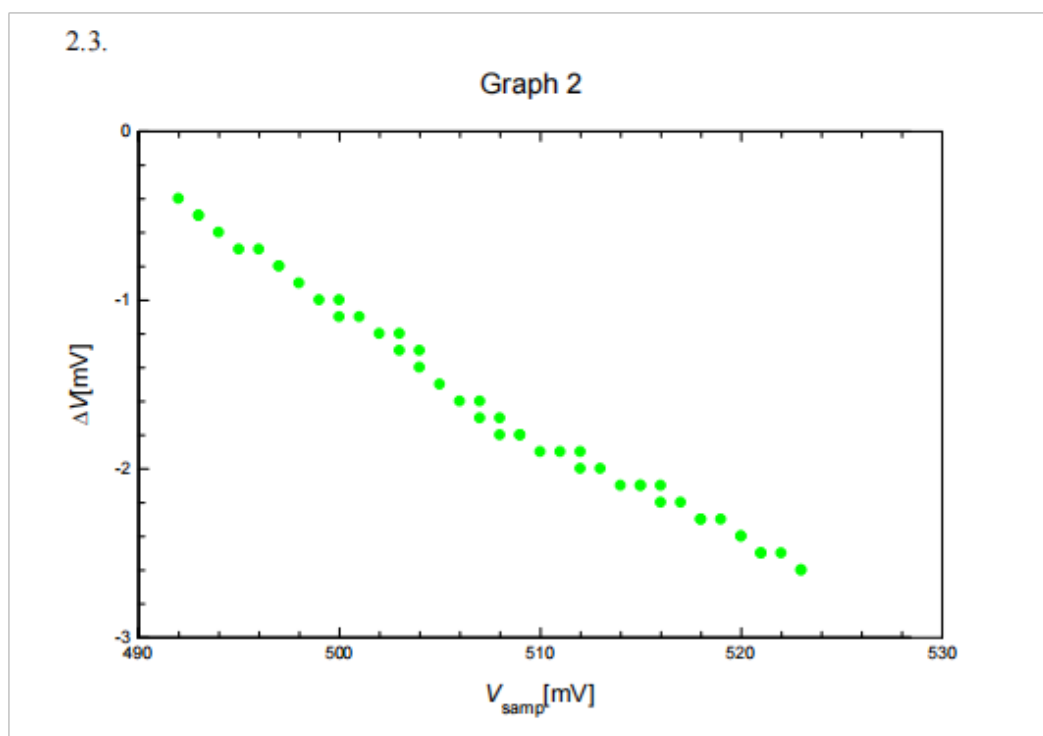
ΔV ning qabul qilinadigan oralig'i ± 40 mV. ΔV ning T bilan o'zgarishi uchun qat'iy qoida yo'q (bu idishlarning plastinkadagi joylashishiga va boshqalarga bog'liq).

2.2.



1-grafik (Graph 1): $V_{\text{samp}} [\text{mV}]$ vs $t [\text{s}]$. To'g'ri grafik qiyalikning hech qanday keskin o'zgarishiga ega bo'lmasligi kerak.

2.3.



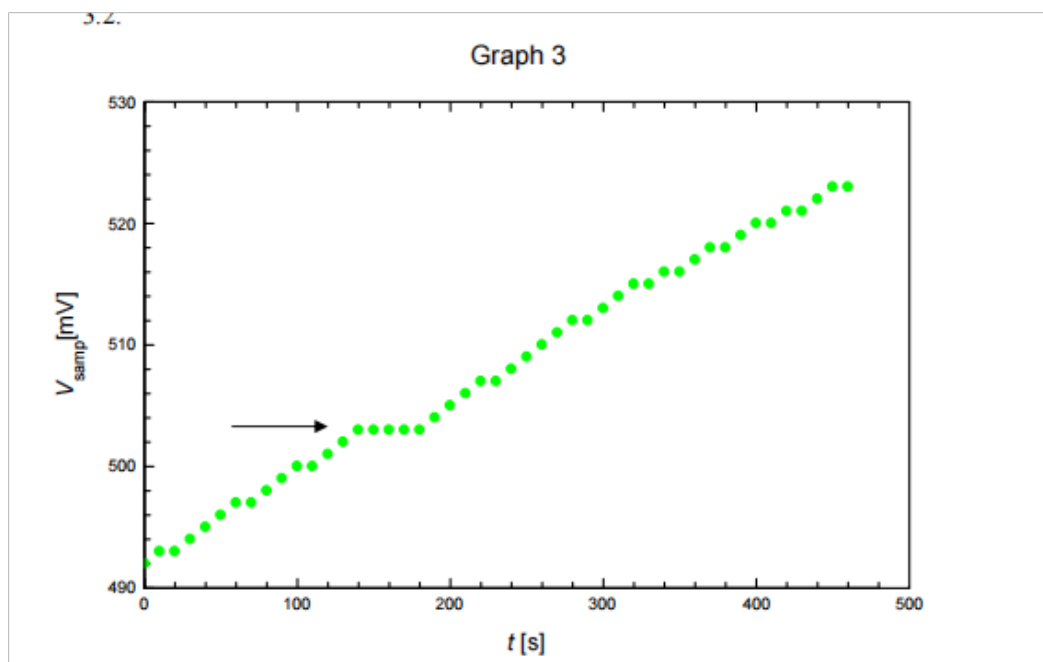
2-grafik (Graph 2): $\Delta V [\text{mV}]$ vs $V_{\text{samp}} [\text{mV}]$. To'g'ri grafik qiyalikning hech qanday keskin o'zgarishiga ega bo'lmasligi kerak.

3.

3.1. Modda solingan idish (Dish with substance)

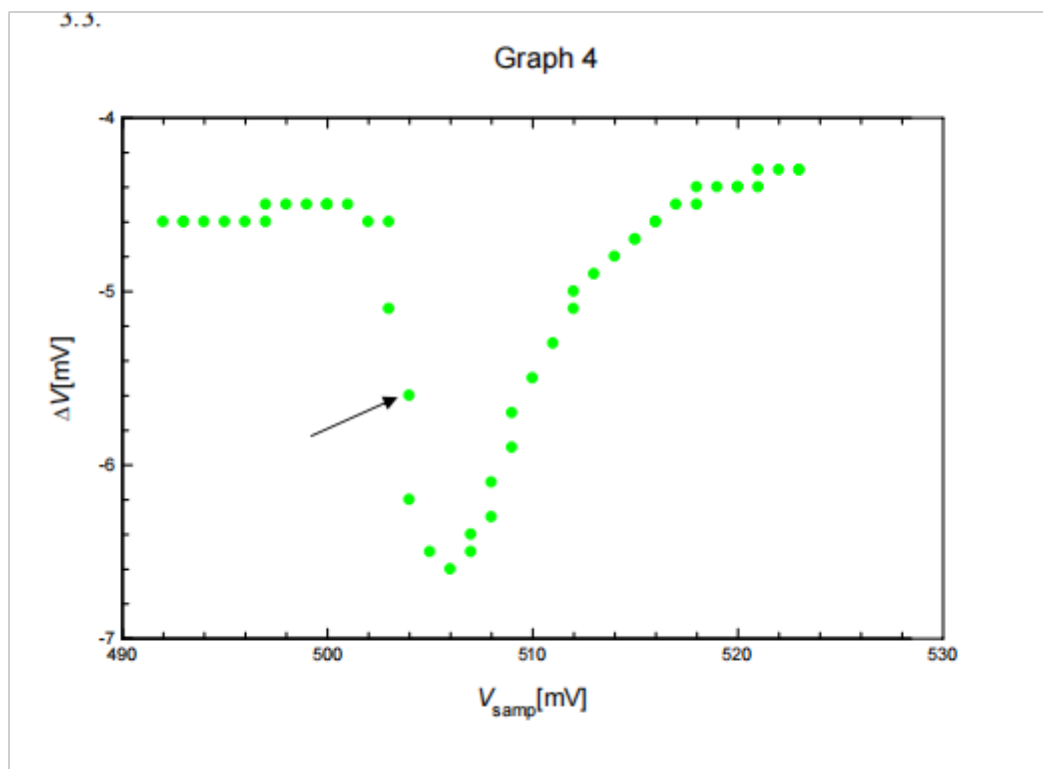
t (s)	V_{samp} (mV) (± 3 mV)	ΔV (mV) (± 0.2 mV)
0	492	-4.6
10	493	-4.6
20	493	-4.6
30	494	-4.6
40	495	-4.6
50	496	-4.6
60	497	-4.6
70	497	-4.5
80	498	-4.5
90	499	-4.5
100	500	-4.5
110	500	-4.5
120	501	-4.5
130	502	-4.6
140	503	-4.6
150	503	-5.1
160	503	-5.6
170	503	-6.2
180	503	-6.5
190	504	-6.6
200	505	-6.5
210	506	-6.4
220	507	-6.3
230	507	-6.1
240	508	-5.9
250	509	-5.7
260	510	-5.5
270	511	-5.3
280	512	-5.1
290	512	-5.0
300	513	-4.9
310	514	-4.8
320	515	-4.7
330	515	-4.7
340	516	-4.6
350	516	-4.6
360	517	-4.5
370	518	-4.5
380	518	-4.4
390	519	-4.4
400	520	-4.4
410	520	-4.4
420	521	-4.4
430	521	-4.3
440	522	-4.3
450	523	-4.3
460	523	-4.3

3.2.



3-grafik (Graph 3): V_{samp} [mV] vs t [s]. To'g'ri 3-grafik yuqoridagi rasmda o'q bilan belgilangan qisqa plato (tekislik) ni o'z ichiga olishi kerak.

3.3.



4-grafik (Graph 4): ΔV [mV] vs V_{samp} [mV].

4-grafik (Graph 4): ΔV [mV] vs V_{samp} [mV]. To'g'ri 4-grafik yuqoridagi rasmda o'q bilan ko'rsatilganidek ΔV da keskin o'zgarishga (abrupt change) ega bo'lishi kerak.

Eslatma (Note): idishda modda bo'lganida, ΔV qiymatlari modda bo'lmagan holatdagiga nisbatan o'zgarishi mumkin.

4.

4.1.

V_s 3-grafikda ko'rsatilgan. $V_s = (503 \pm 3)$ mV. Bundan $T_s = 60.5^\circ\text{C}$ chiqarilishi mumkin.

4.2.

V_s 4-grafikda ko'rsatilgan. $V_s = (503 \pm 3)$ mV. Bundan $T_s = 60.5^\circ\text{C}$ chiqarilishi mumkin.

4.3.

O'rtacha kvadratik usul (root mean square method) yordamida xatolik hisoblari (Error calculations):

$$T_s = T_0 + \frac{V(T_0) - V(T_s)}{\alpha} = T_0 + A, \text{ bu yerda } A - \text{oraliq o'zgaruvchi.}$$

Shuning uchun T_s ning xatoligi $\delta T_s = \sqrt{(\delta T_0)^2 + (\delta A)^2}$ ko'rinishida yozilishi mumkin, bu yerda $\delta \dots$ – xatolik.

A uchun xatolik alohida hisoblanadi:

$$\delta A = \frac{V(T_0) - V(T_s)}{\alpha} \sqrt{\left[\frac{\delta[V(T_0) - V(T_s)]}{V(T_0) - V(T_s)} \right]^2 + \left(\frac{\delta \alpha}{\alpha} \right)^2}$$

bu yerda:

$$\delta[V(T_0) - V(T_s)] = \sqrt{[\delta V(T_0)]^2 + [\delta V(T_s)]^2}$$

Ushbu tajribadagi boshqa o'zgaruvchilarning xatoliklari (Errors of other variables):

- $\delta T_0 = 1^\circ\text{C}$
- $\delta V(T_0) = 3$ mV, multimetrdan o'qilgan.
- $\delta \alpha = 0.03$ mV/ $^\circ\text{C}$
- $\delta V(T_s) \approx 3$ mV

Yuqoridagi tashkil etuvchi xatoliklardan:

$$\delta[V(T_0) - V(T_s)] \approx 4.24 \text{ mV}$$

Nihoyat, T_s ning xatoligi: $\delta T_s \approx 2.5^\circ\text{C}$

Shunday qilib, yakuniy natija: $T_s = 60 \pm 2.5^\circ\text{C}$

Eslatma: Agar talaba taxminan bir xil natijaga olib keladigan boshqa har qanday asosli xatolik hisoblash usulidan foydalansa, u ham qabul qilinadi.

2-topshiriq (Task 2)

1.

1.1. $T_0 = 26 \pm 1^\circ\text{C}$

2.

2.1. Lampa o'chirilgan holda o'lchangan ma'lumotlar (Measured data with the lamp off)

t (s)	$\Delta V(T_0)$ (mV) (± 0.2 mV)
0	19.0
10	19.0
20	19.0
30	19.0
40	19.0
50	18.9
60	18.9
70	18.9
80	18.9
90	18.9
100	19.0
110	19.0
120	19.0

$\Delta V(T_0)$ qiymatlari bir tajriba to'plamidan boshqasiga farq qilishi mumkin. Qabul qilinadigan qiymatlar -40 dan $+40$ mV gacha oralig'ida yotadi.

2.2. Lampa yoqilgan holda o'lchangan ma'lumotlar (Measured data with the lamp on)

t (s)	ΔV (mV) (± 0.2 mV)
0	19.5
10	21.9
20	23.8
30	25.5
40	26.9
50	28.0
60	29.0
70	29.9
80	30.7
90	31.4

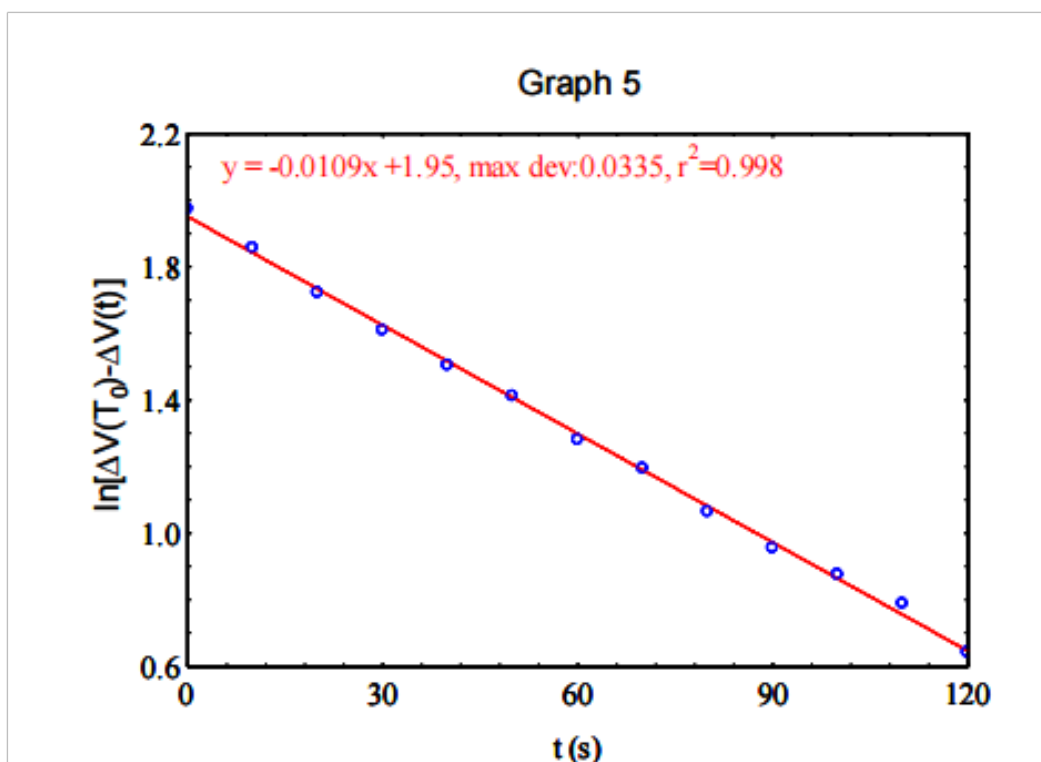
t (s)	ΔV (mV) (± 0.2 mV)
0	23.2
10	22.4
20	21.6
30	21.0
40	20.5
50	20.1
60	19.6
70	19.3
80	18.9
90	18.6
100	18.4
110	18.2
120	17.9

3. 5-grafikni chizish va k ni hisoblash (Plotting graph 5 and calculating k)

3.1. $x = t$; $y = \ln [\Delta V(T_0) - \Delta V(t)]$

Eslatma: In yordamida chiziqli munosabatga olib keladigan x va y uchun ifodalarni yozishning boshqa asosli usullari ham qabul qilinadi.

3.2.



5-grafik (Graph 5).

$$y = -0.0109x + 1.95, \quad \text{maksimal og'ish (max dev): } 0.0335, \quad r^2 = 0.998$$

3.3. k ni hisoblash (Calculating k):

qiyalik (slope) $= \frac{k}{C} = 0.0109 \text{ s}^{-1}$ va $C = 0.69 \text{ J/K}$, shuning uchun: $k = 7.52 \times 10^{-3} \text{ W/K}$

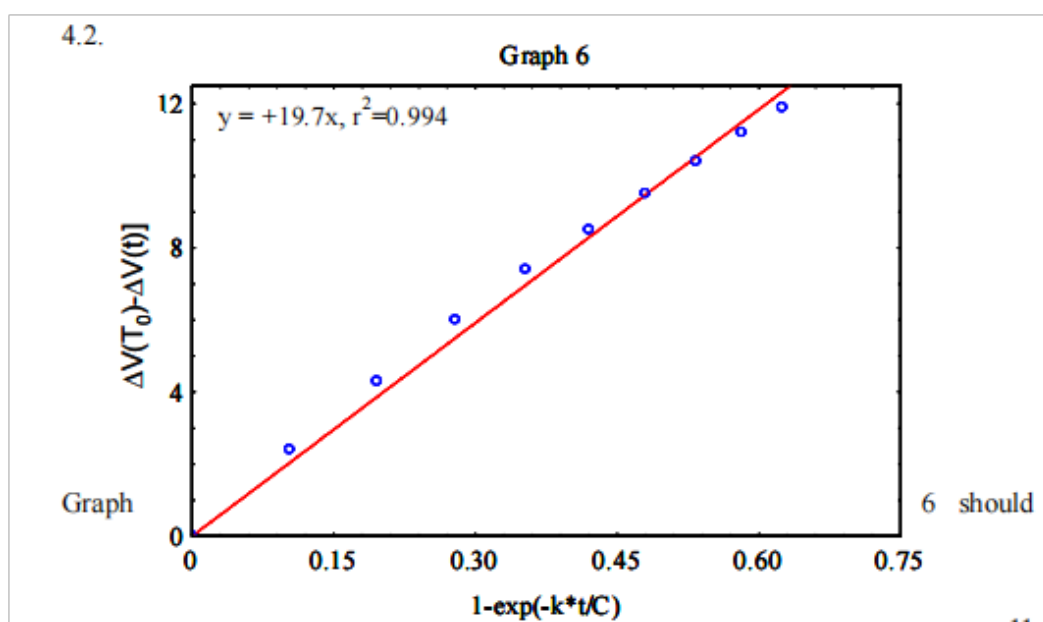
Eslatma: k ning xatoligi 5.5 da hisoblanadi. Talabalardan ushbu bosqichda k ning xatoligini berish so'ralmaydi. Tajriba to'plamiga qarab k ning qabul qilinadigan qiymati $6 \times 10^{-3} \div 9 \times 10^{-3} \text{ W/K}$ oralig'ida yotadi.

4. 6-grafikni chizish va E ni hisoblash (Plotting Graph 6 and calculating E)

4.1.

$$x = \left[1 - \exp\left(\frac{-kt}{C}\right) \right]; \quad y = |\Delta V(T_0) - \Delta V(t)|$$

4.2.



6-grafik (Graph 6).

$$y = +19.7x, \quad r^2 = 0.994$$

4.3. (davomi)

4.3. 6-grafikning qiyaligidan (slope) va detektor teshigining maydonidan (area of the detector orifice) biz $E = 140 \text{ W/m}^2$ ni olamiz. Detektor teshigining maydoni:

$$S_{\text{det}} = \pi R_{\text{det}}^2 = \pi \times (13 \times 10^{-3})^2 = 5.30 \times 10^{-4} \text{ m}^2$$

xatolik bilan (with error):

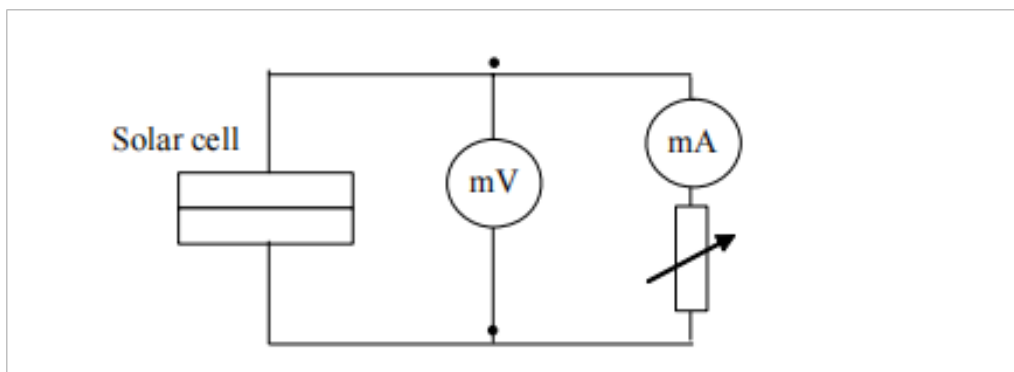
$$\frac{\delta R_{\text{det}}}{R_{\text{det}}} = 5\%$$

E ning xatoligi 5.5 da hisoblanadi. Talabalardan ushbu bosqichda E ning xatoligini berish so'ralmaydi. Tajriba to'plamiga qarab E ning qabul qilinadigan qiymati $120 \div 160 \text{ W/m}^2$ oralig'ida yotadi.

5.

5.1.

Zanjir diagrammasi (Circuit diagram):



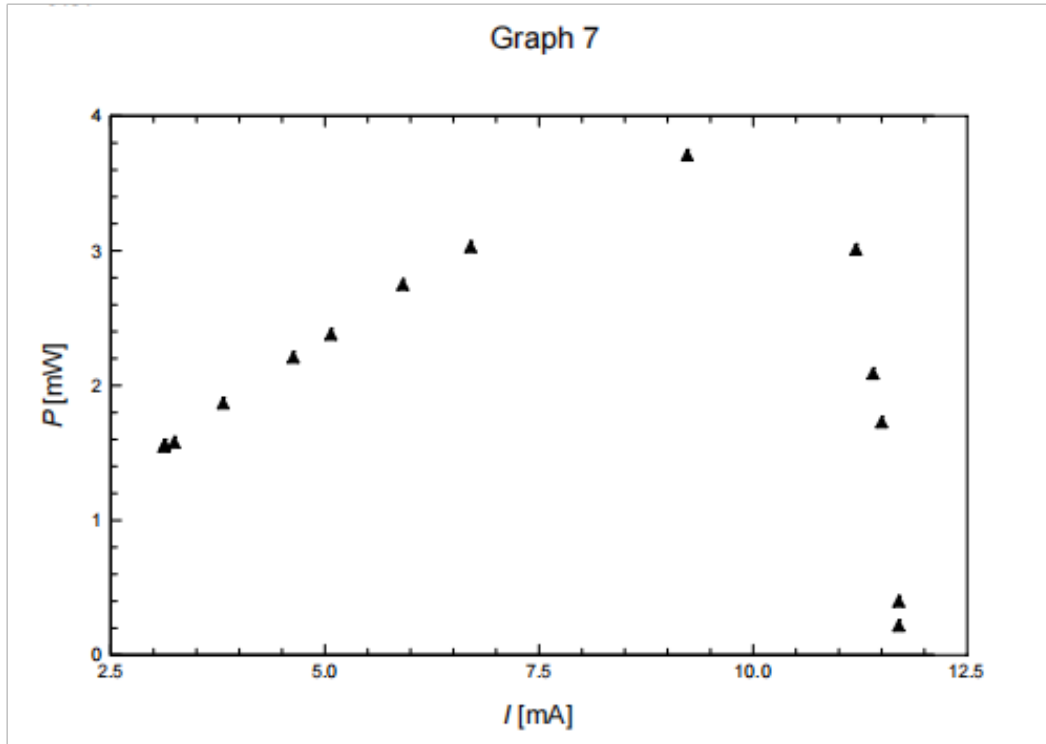
Zanjir diagrammasi (Circuit diagram).

5.2.

V va I ning o'lchovlari (Measurements of V and I)

V (mV) ($\pm 0.3 \div \pm 3$ mV)	I (mA) ($\pm 0.05 \div \pm 0.1$ mA)	P (mW)
18.6 $\pm$ 0.3	11.7	0.21
33.5	11.7	0.39
150	11.5	1.72
157	11.6	1.82
182 $\pm$ 1	11.4	2.08
267	11.2	3.00
402 $\pm$ 2	9.23	3.70
448	6.70	3.02
459	5.91	2.74
468	5.07	2.37
473 $\pm$ 3	4.63	2.20
480	3.81	1.86
485	3.24	1.57
487	3.12	1.54
489	3.13	1.55

5.3.



7-grafik (Graph 7).

5.4.

$$P_{\max} = 3.7 \pm 0.2 \text{ mW}$$

Tajriba to'plamiga qarab $P_{\max}$ ning qabul qilinadigan qiymati $3 \div 4.5 \text{ mW}$ oralig'ida yotadi.

5.5.

Samaradorlik (efficiency) uchun ifoda (Expression for the efficiency)

$$S_{\text{cell}} = 19 \times 24 \text{ mm}^2 = 450 \times 10^{-6} \text{ m}^2$$

Keyin (Then)

$$\eta_{\max} = \frac{P_{\max}}{E \times S_{\text{cell}}} = 0.058$$

Xatolik hisobi (Error calculation):

$$\delta\eta_{\max} = \eta_{\max} \sqrt{\left(\frac{\delta P_{\max}}{P_{\max}}\right)^2 + \left(\frac{\delta E}{E}\right)^2 + \left(\frac{\delta S_{\text{cell}}}{S_{\text{cell}}}\right)^2}, \text{ bu yerda } S_{\text{cell}} - \text{quyosh batareyasining maydoni (area)}$$

$$\frac{\delta P_{\max}}{P_{\max}} \text{ 7-grafikdan baholanadi, odatdagi qiymat } \approx 6\%$$

$$\frac{\delta S_{\text{cell}}}{S_{\text{cell}}} : \text{ millimetr olchovidan (chizgich bilan) xatolik, odatdagi qiymat } \approx 5\%$$

E o'rtacha nisbatdan hisoblanadi (6-grafik yordamida) (E is calculated from averaging the ratio (using Graph 6)):

$$B = \frac{\Delta V(T_0) - \Delta V(t)}{1 - \exp\left(-\frac{k}{C}t\right)} = \frac{E\pi R_{\text{det}}^2 \alpha}{k}$$

bu yerda B – oraliq o'zgaruvchi (intermediate variable), R_{det} – detektor teshigining radiusi (radius of the detector orifice).

$$E = \frac{kB}{\pi R_{\text{det}}^2 \alpha}$$

E ning xatoligini hisoblash (Calculation of error of E):

$$\left(\frac{\delta E}{E}\right) = \sqrt{\left(\frac{\delta k}{k}\right)^2 + \left(\frac{\delta B}{B}\right)^2 + 4\left(\frac{\delta R_{\text{det}}}{R_{\text{det}}}\right)^2 + \left(\frac{\delta \alpha}{\alpha}\right)^2}$$

k quyidagi regressiyadan hisoblanadi (k is calculated from the regression of):

$$\Delta T = \Delta T(0) \exp\left(-\frac{k}{C}t\right), \text{ shuning uchun } \ln \Delta T = \ln \Delta T(0) - \frac{k}{C}t$$

Biz $k/C = m$ deb belgilaymiz, keyin $k = mC$.

Regressiyadan m ning xatoligini hisoblash mumkin (From the regression, we can calculate the error of m):

$$\frac{\delta m}{m} \approx 2(1 - r) \approx 0.2\%$$

$$\frac{\delta k}{k} = \sqrt{\left(\frac{\delta m}{m}\right)^2 + \left(\frac{\delta C}{C}\right)^2}$$

Biz η_{max} xatoligi uchun ifodani chiqaramiz (We derive the expression for the error of η_{max}):

$$\delta \eta_{\text{max}} = \eta_{\text{max}} \sqrt{\left(\frac{\delta P_{\text{max}}}{P_{\text{max}}}\right)^2 + \left(\frac{\delta S_{\text{cell}}}{S_{\text{cell}}}\right)^2 + \left(\frac{\delta B}{B}\right)^2 + 4\left(\frac{\delta R_{\text{det}}}{R_{\text{det}}}\right)^2 + \left(\frac{\delta m}{m}\right)^2 + \left(\frac{\delta C}{C}\right)^2 + \left(\frac{\delta \alpha}{\alpha}\right)^2}$$

η_{max} va boshqa tashkil etuvchi xatoliklar uchun odatdagi qiymatlar (Typical values for η_{max} and other constituent errors):

$$\eta_{\text{max}} \approx 0.058$$

$$\frac{\delta P_{\text{max}}}{P_{\text{max}}} = 5\% \quad ; \quad \frac{\delta B}{B} \approx 0.6\% \quad ; \quad \frac{\delta m}{m} \approx 0.2\% \quad ; \quad \frac{\delta S_{\text{cell}}}{S_{\text{cell}}} \approx 5\% \quad ; \quad \frac{\delta R_{\text{det}}}{R_{\text{det}}} \approx 5\% \quad ; \quad \frac{\delta C}{C} \approx 3\% \quad ; \quad \frac{\delta \alpha}{\alpha} \approx 0.1\%$$

Nihoyat (Finally):

$$\frac{\delta \eta_{\text{max}}}{\eta_{\text{max}}} = 12.7\%, \quad \delta \eta_{\text{max}} \approx 0.0074$$

va (and)

$$\eta_{\max} = (5.8 \pm 0.8)\%$$

Eslatma: Agar talaba taxminan bir xil natijaga olib keladigan boshqa har qanday asosli xatolik usulidan foydalansa, u ham qabul qilinadi.